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Tsuji

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(54) **METHOD FOR MANUFACTURING SEMICONDUCTOR DEVICE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 452 days.

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Primary Examiner — Jose R Diaz

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(74) *Attorney, Agent, or Firm* — IPUSA, PLLC

(30) **Foreign Application Priority Data**

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(57) **ABSTRACT**

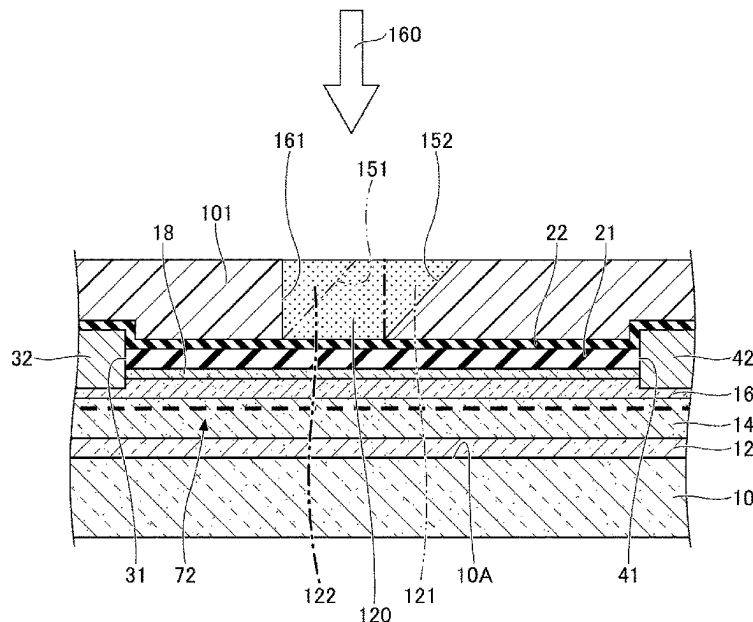
(51) **Int. Cl.**
H10D 30/01 (2025.01)
G03F 7/20 (2006.01)
(Continued)

A method for manufacturing a semiconductor device, includes forming source and drain electrodes on a semiconductor layer provided above a substrate; forming a first insulating film covering a surface of the semiconductor layer, between the source and drain electrodes, forming a second insulating film on the first insulating film, forming a mask on the second insulating film, the mask having an opening between the source and drain electrodes in a plan view viewed in a direction perpendicular to a substrate surface, forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first and second insulating films through the opening, and forming a gate electrode on the first and second insulating films, the gate electrode making a Schottky contact with the semiconductor layer through the first and second gate openings.

(52) **U.S. Cl.**
CPC **H10D 30/015** (2025.01); **G03F 7/201** (2013.01); **G03F 7/203** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC H01L 21/31144; H01L 27/1225; H10D 64/68; H10D 64/681; H10D 64/683;
(Continued)

15 Claims, 33 Drawing Sheets



- (51) **Int. Cl.**
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H01L 21/027 (2006.01)
H01L 21/311 (2006.01)
H10D 30/47 (2025.01)
H10D 64/01 (2025.01)
H10D 64/27 (2025.01)
- (52) **U.S. Cl.**
CPC *H01L 21/0217* (2013.01); *H01L 21/0271*
(2013.01); *H01L 21/0274* (2013.01); *H01L*
21/31144 (2013.01); *H10D 30/061* (2025.01);
H10D 30/4755 (2025.01); *H10D 64/01*
(2025.01); *H10D 64/411* (2025.01)
- (58) **Field of Classification Search**
CPC H10D 64/64; H10D 30/6738; H10D
30/47–476; H10D 30/0015; H10D
86/423; H10D 30/015; G03F 7/201
See application file for complete search history.

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FIG.1

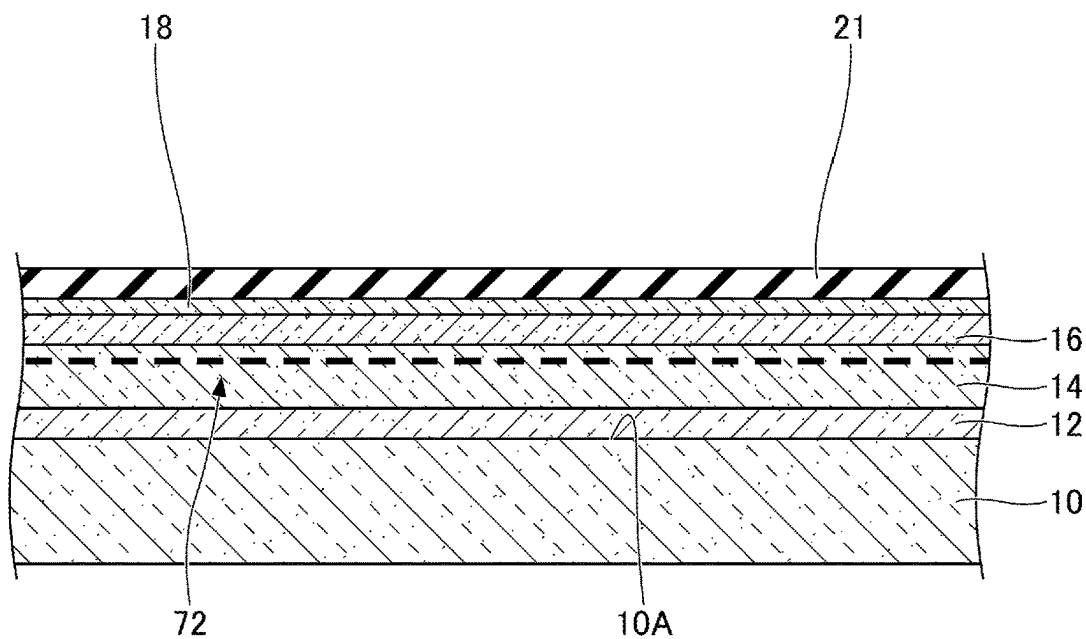


FIG.2

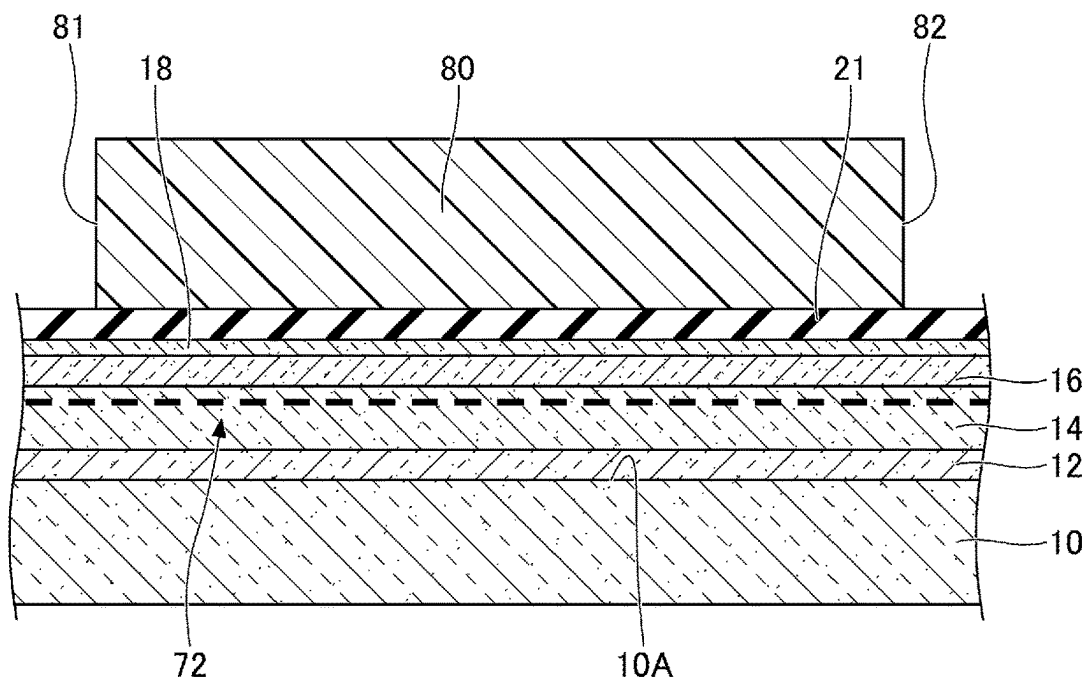


FIG.3

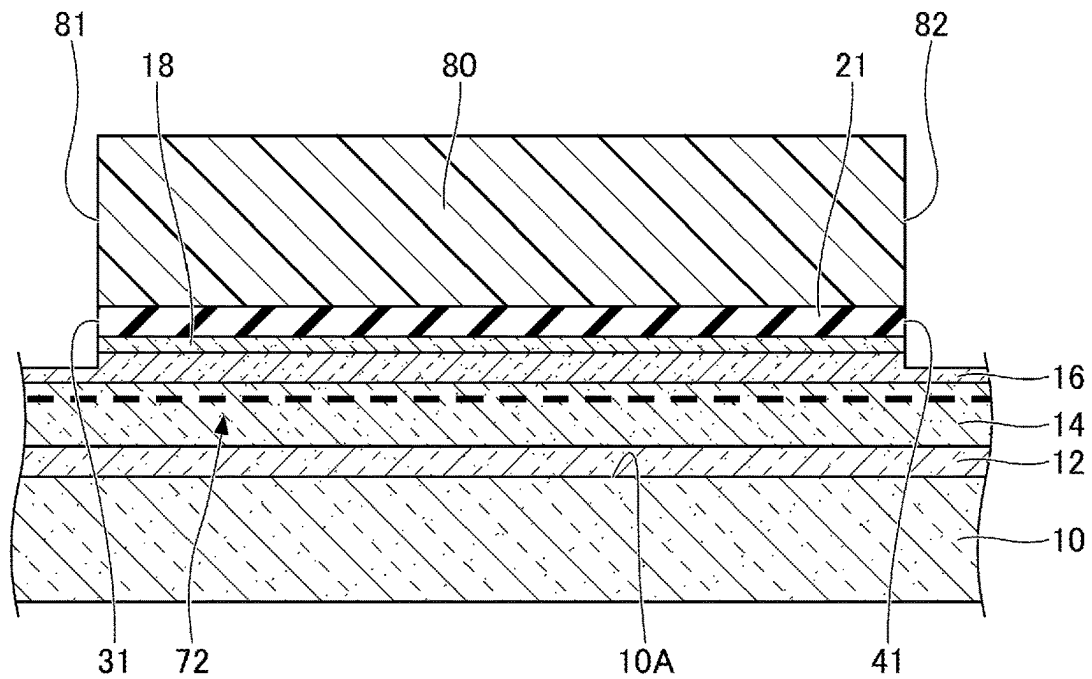


FIG.4

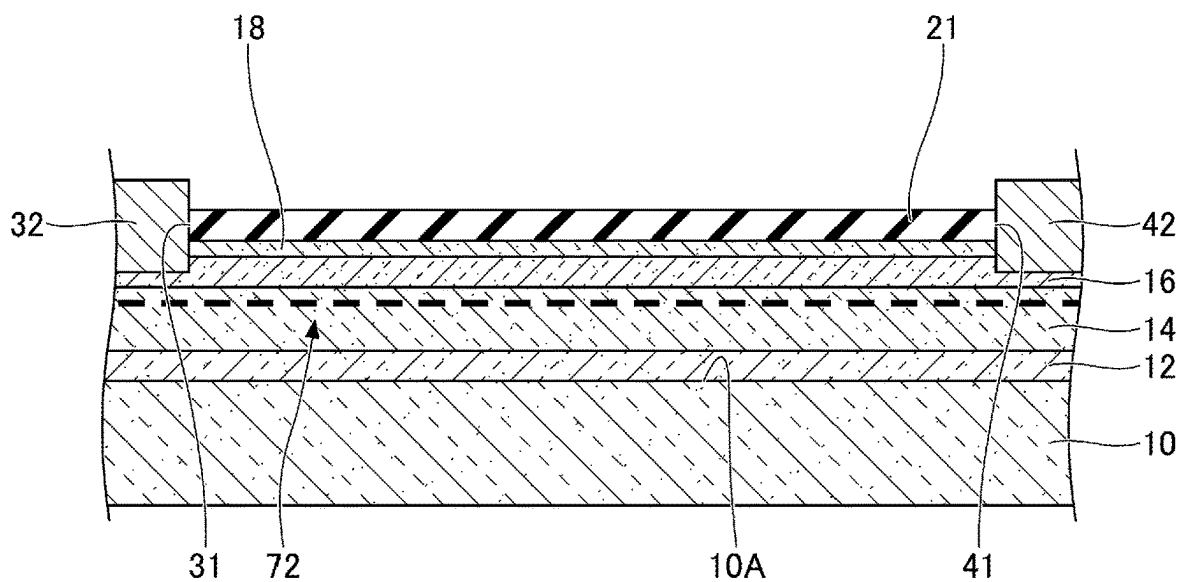


FIG.5

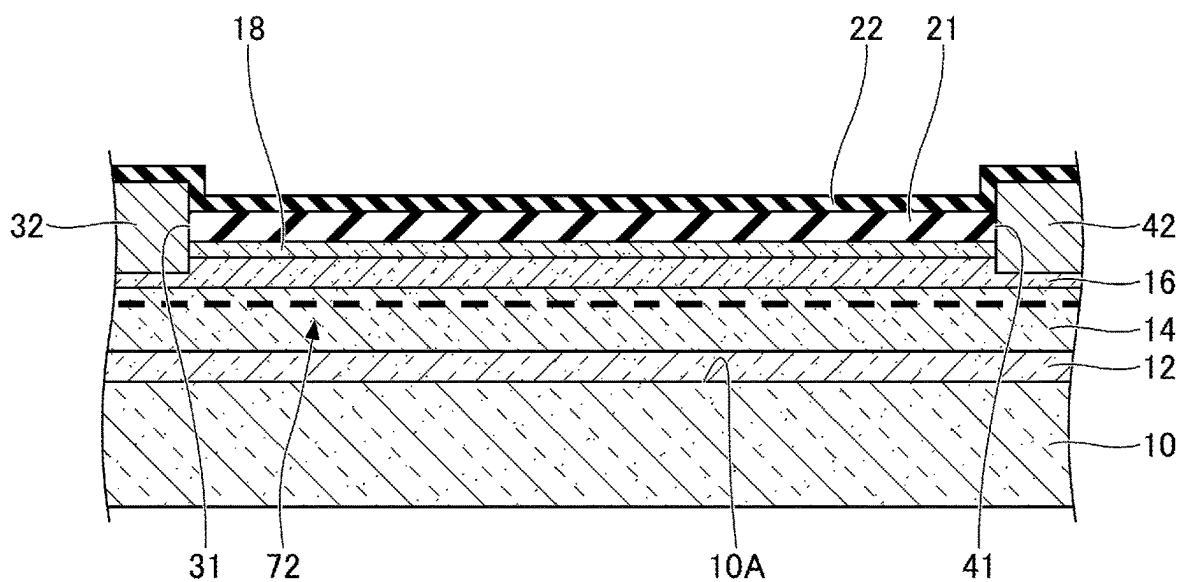


FIG.6

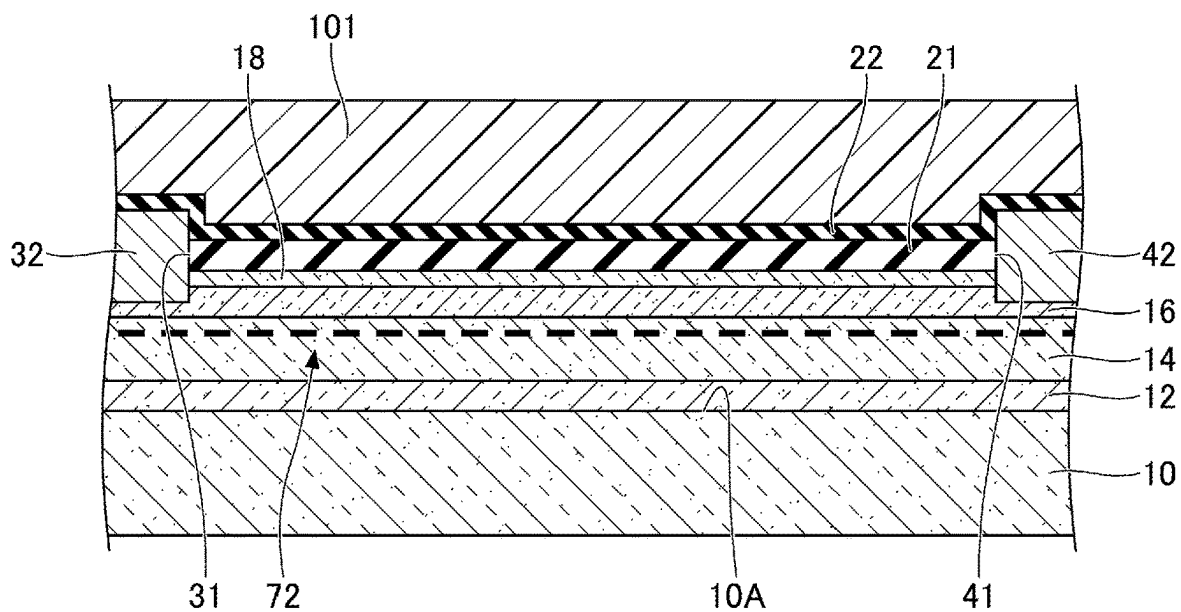


FIG. 7

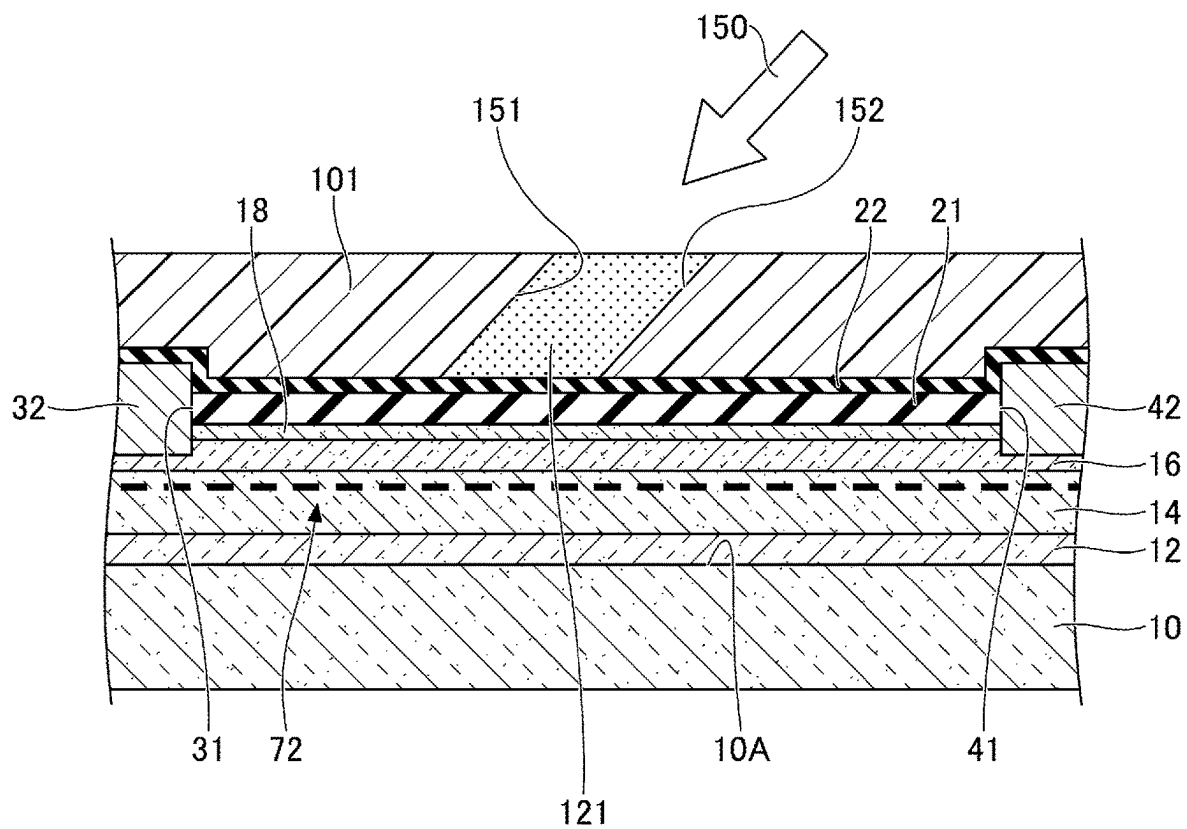


FIG.8

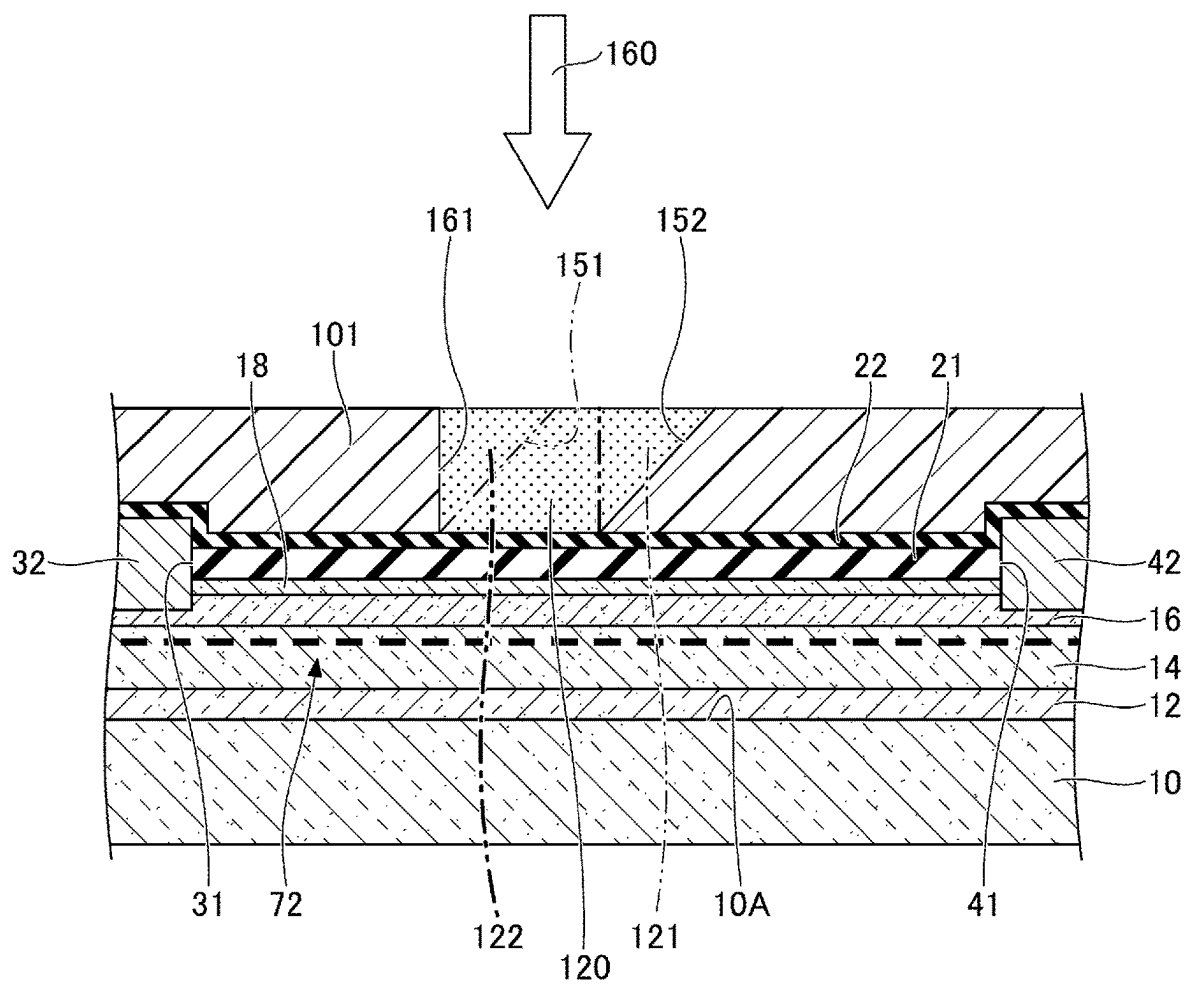
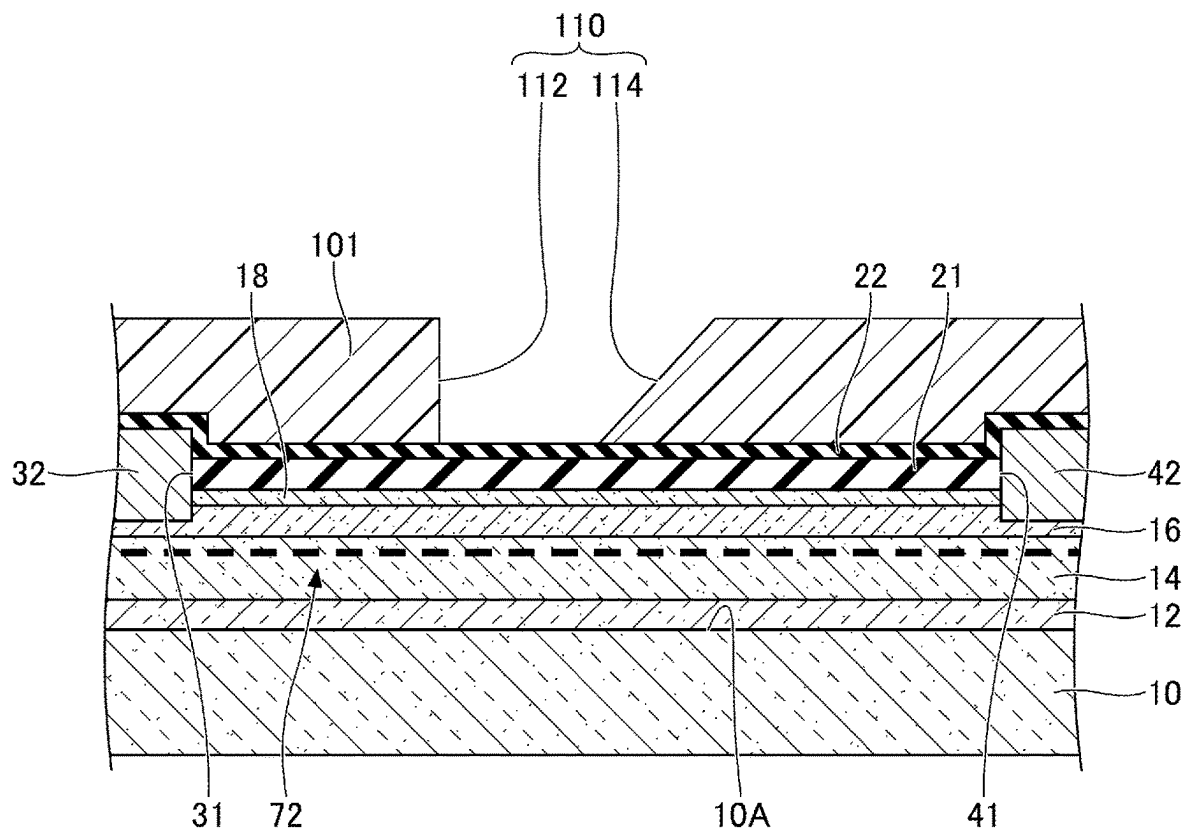


FIG.9



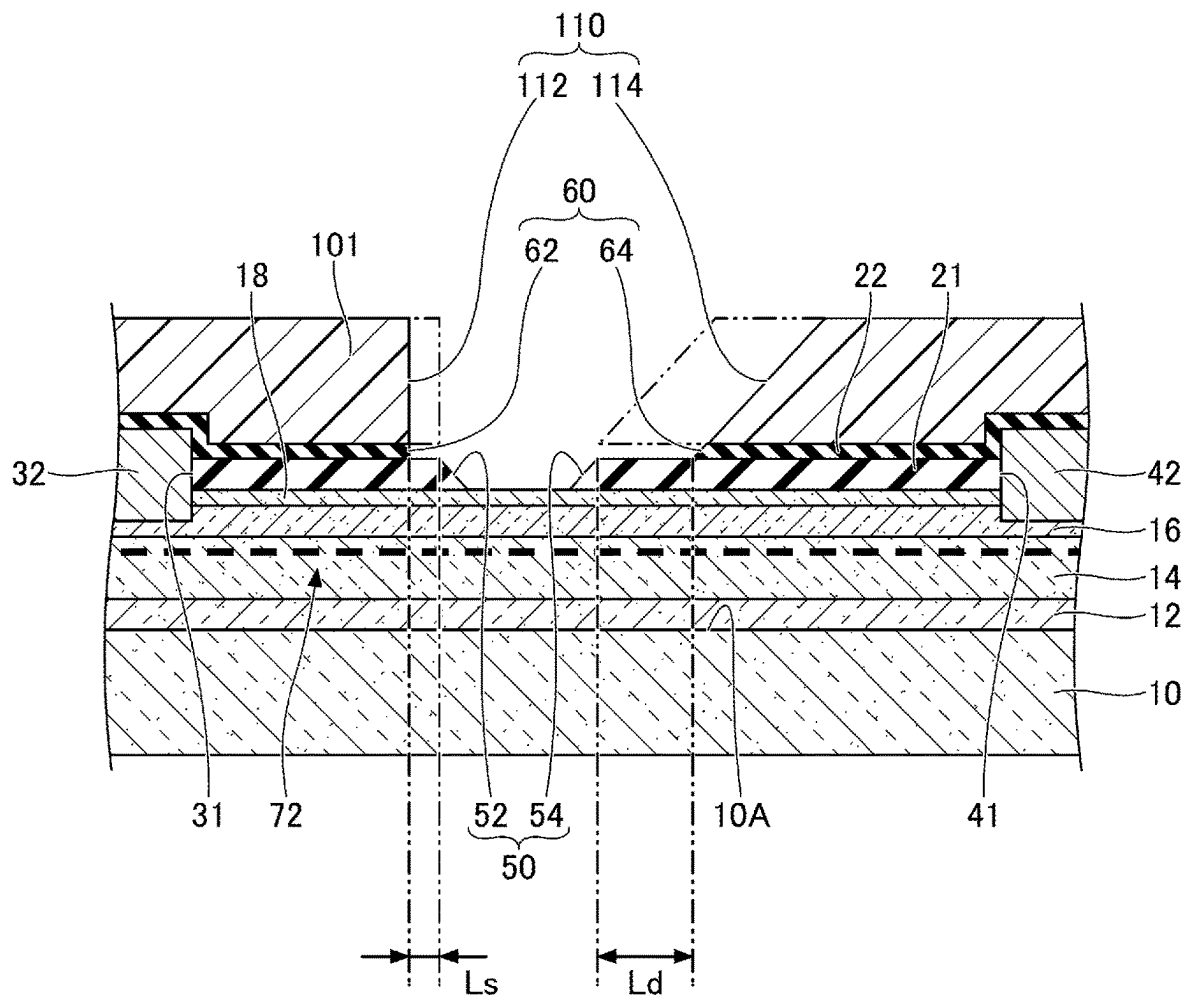


FIG. 11

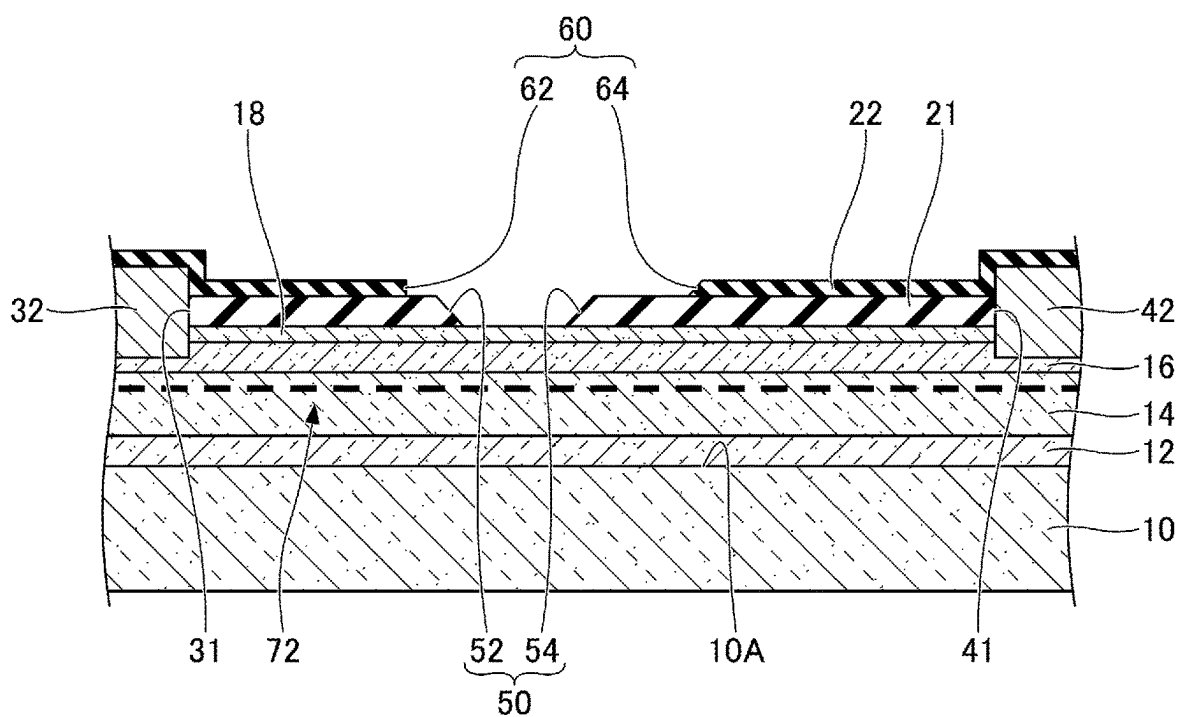


FIG. 12

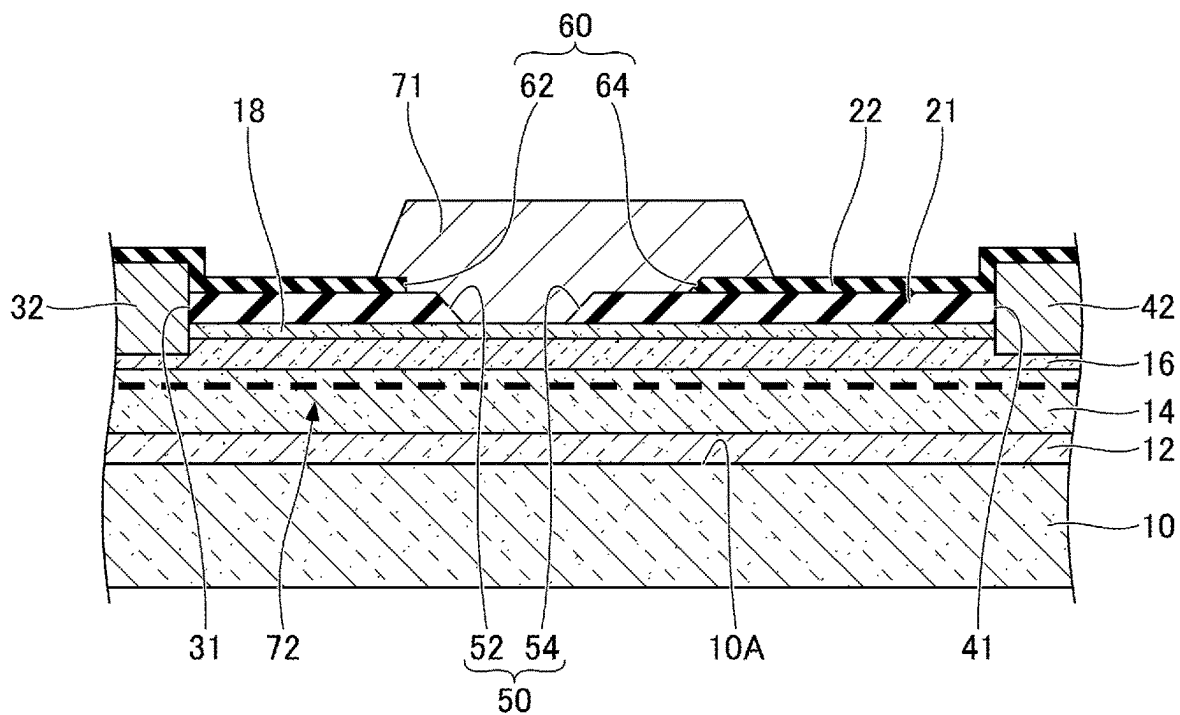


FIG.13

100

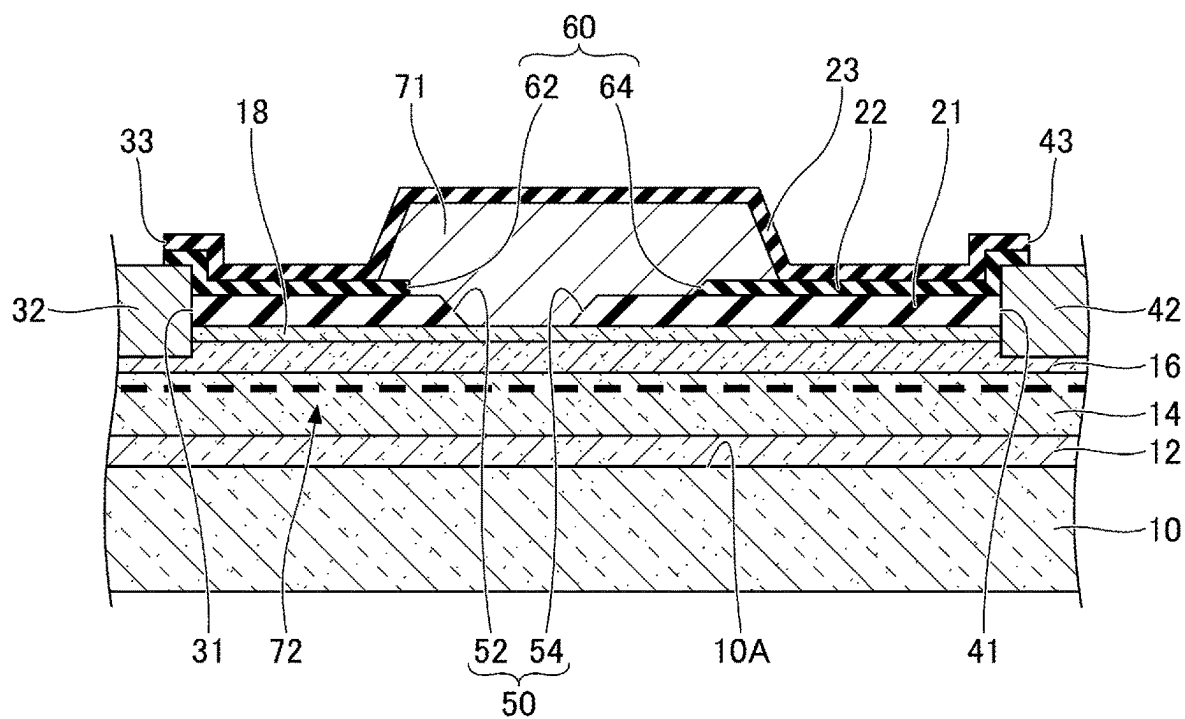


FIG. 14

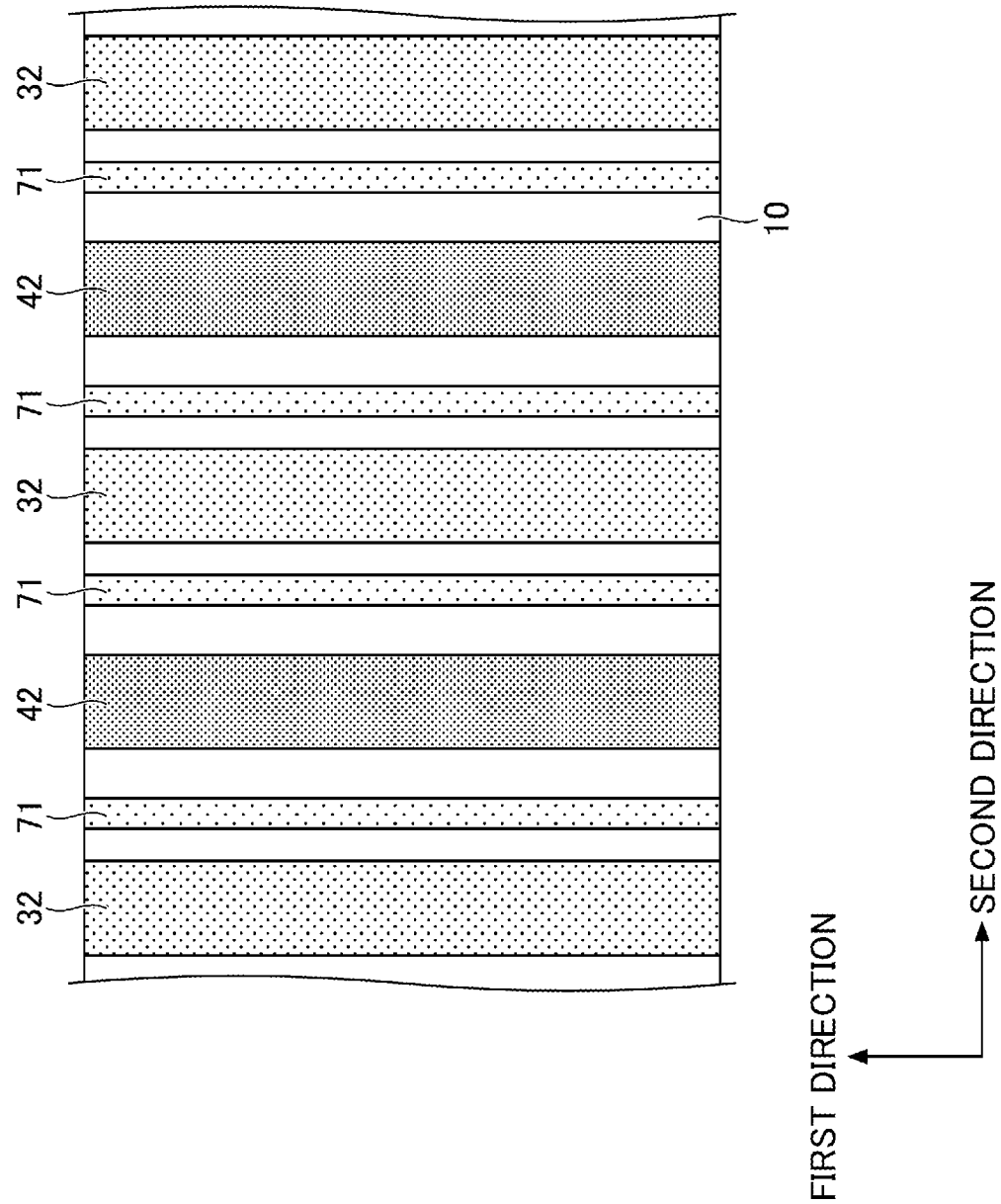


FIG.15

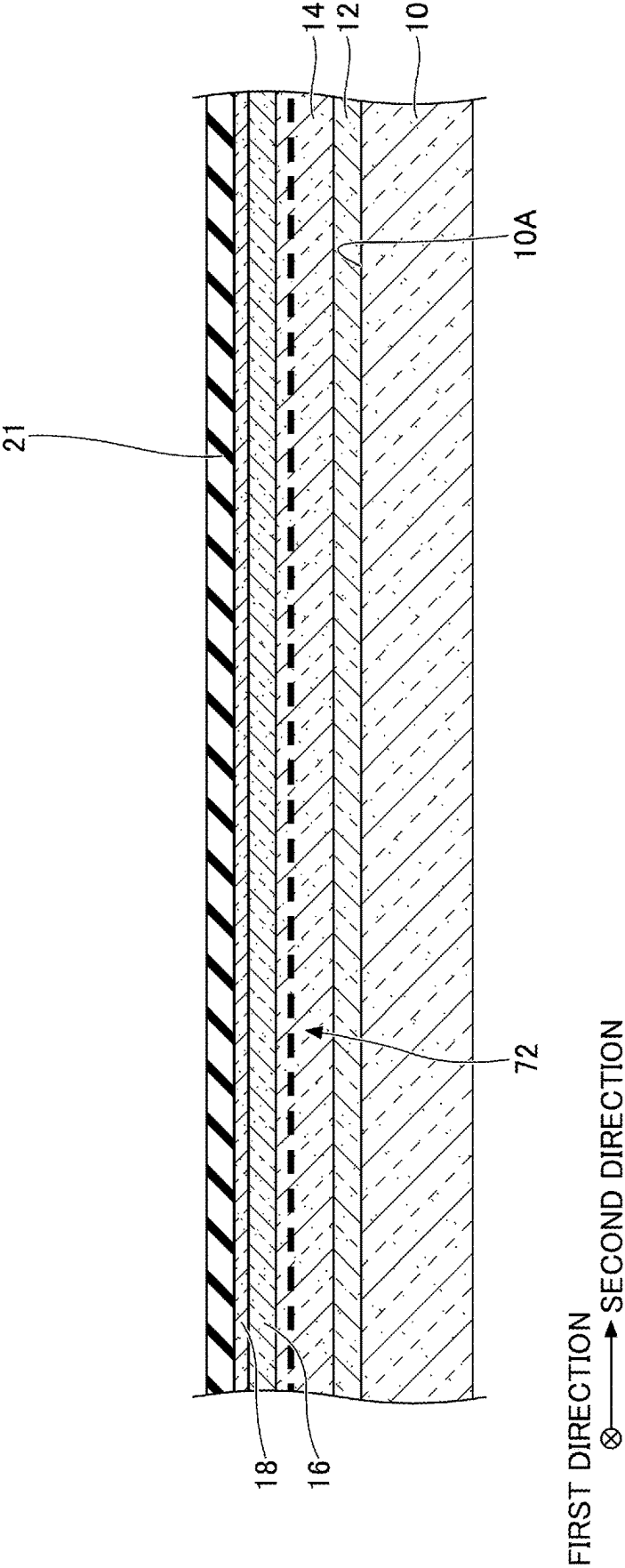


FIG. 16

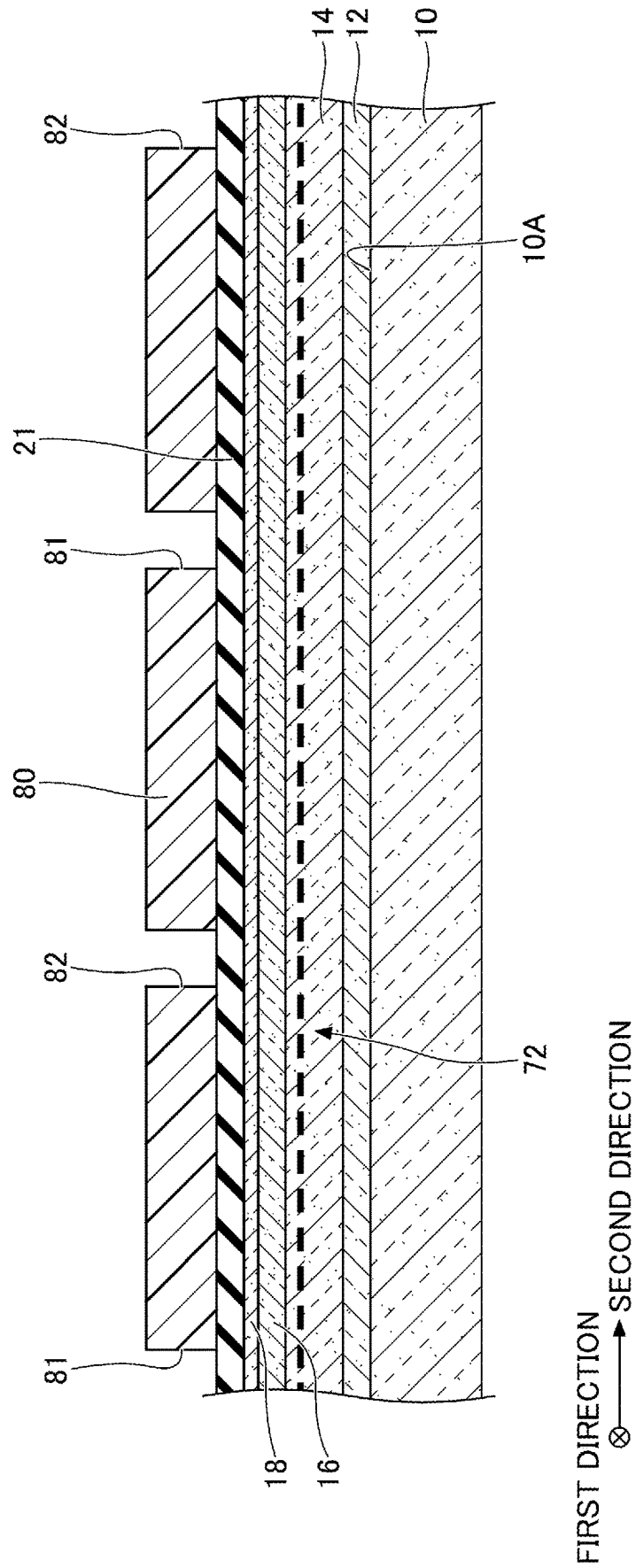
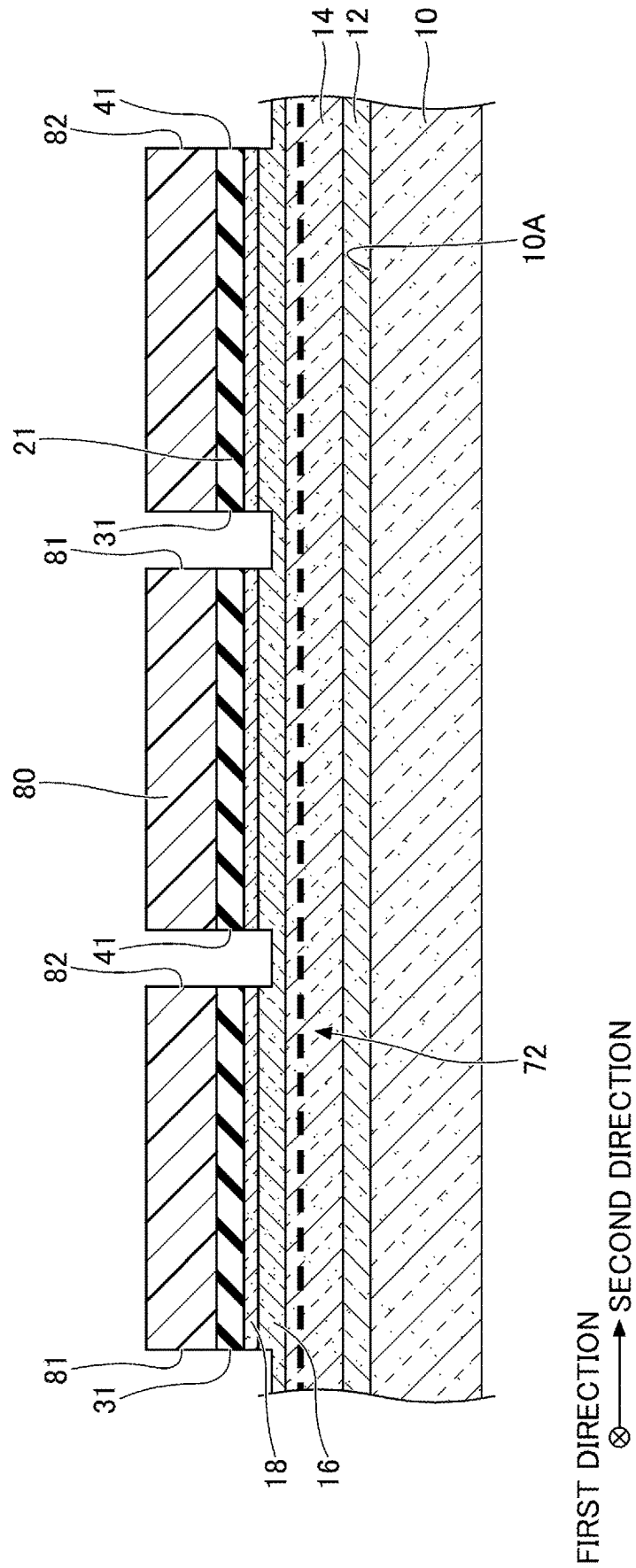


FIG. 17



8
1
G
E

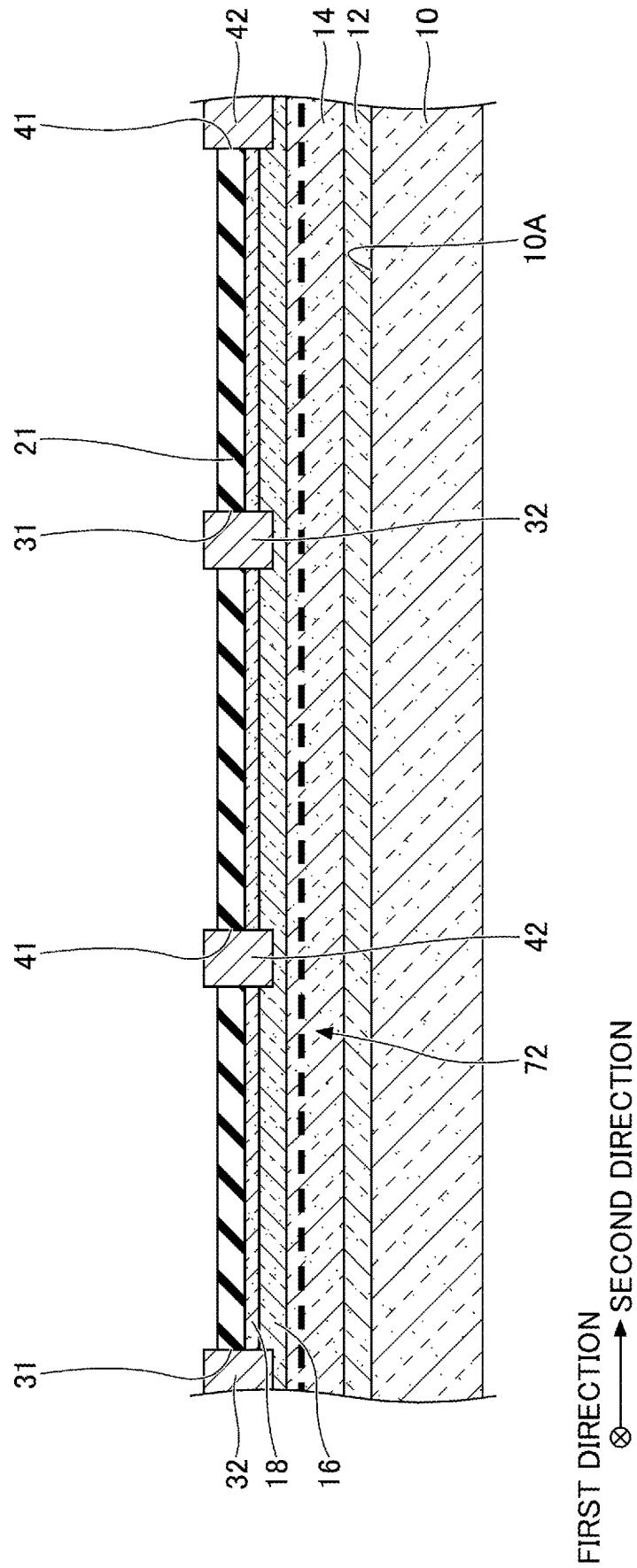


FIG. 19

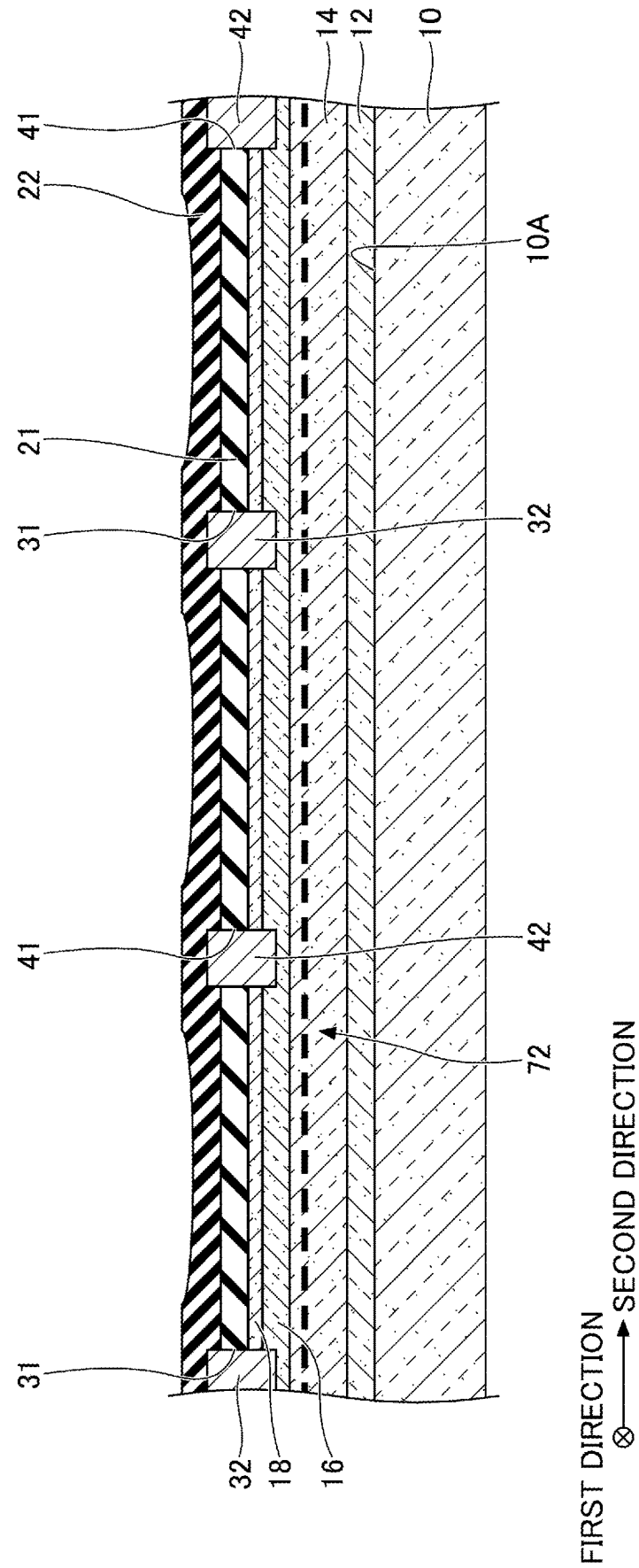


FIG. 20

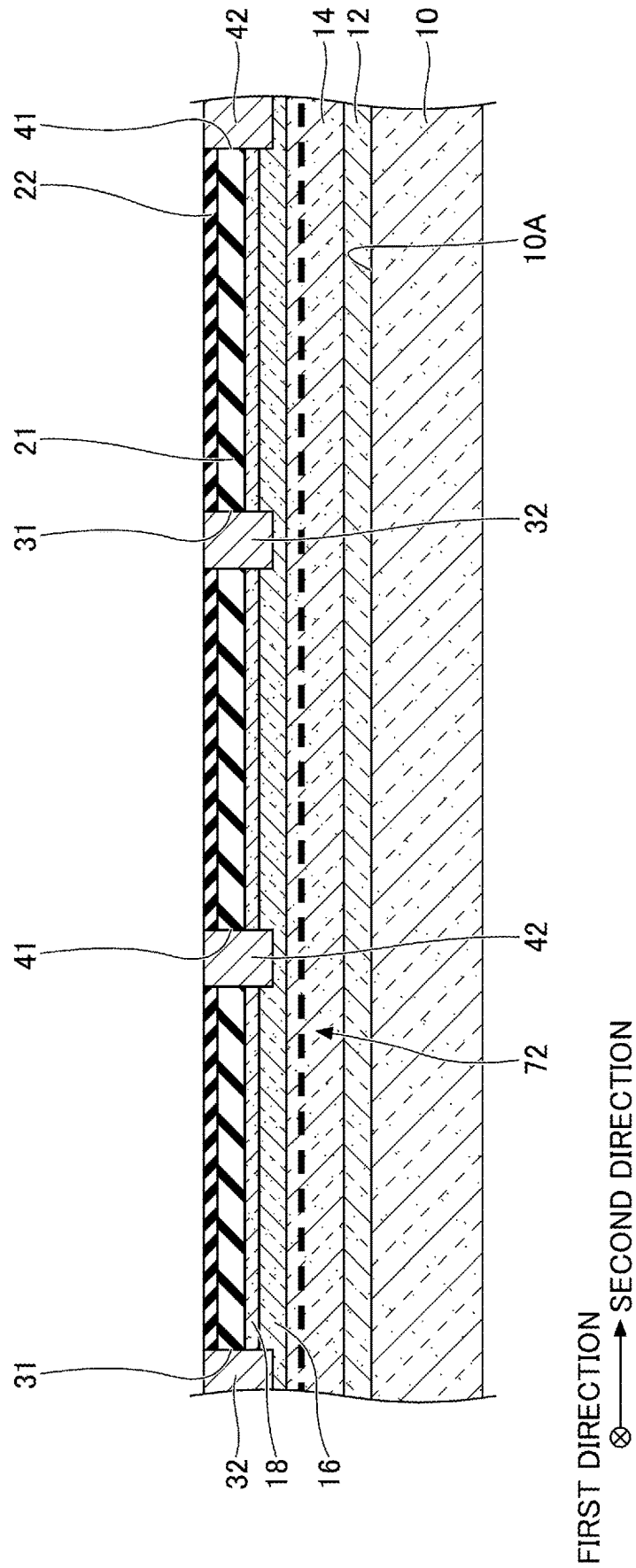


FIG. 21

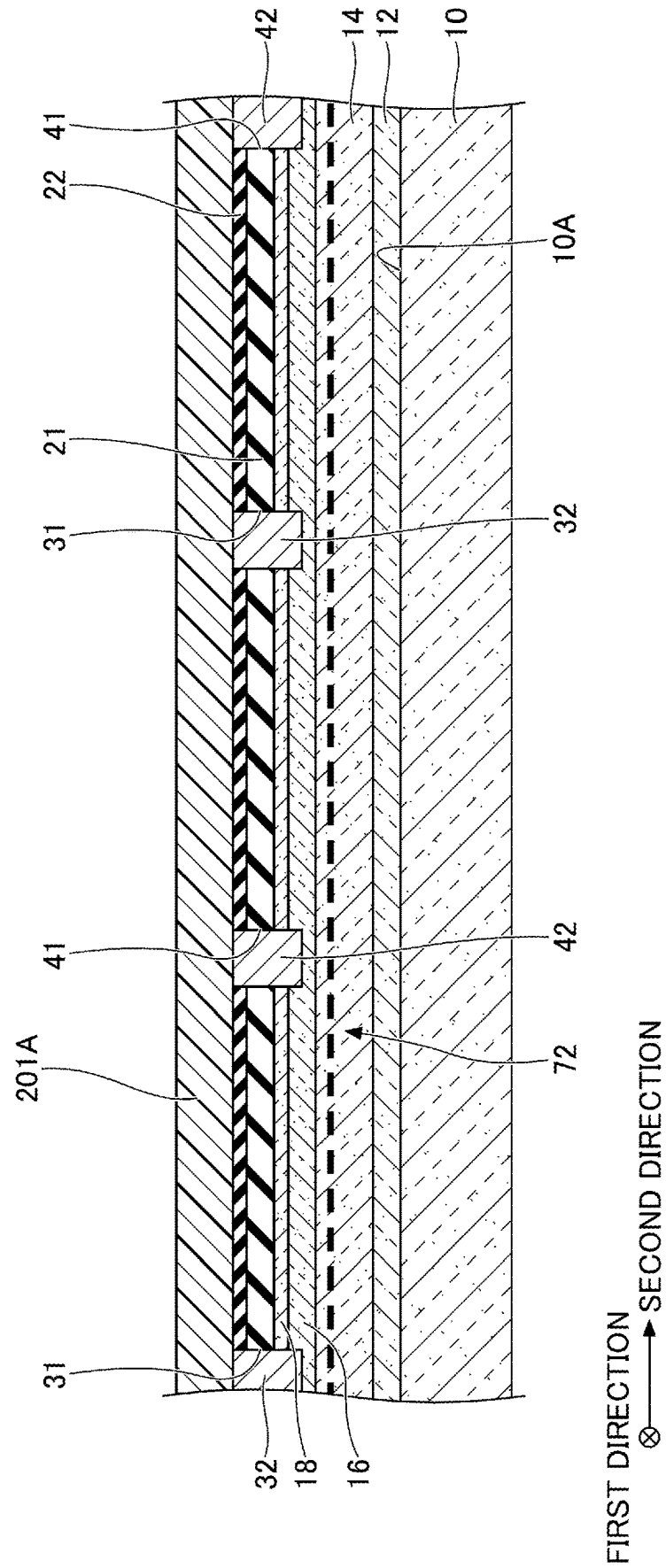


FIG.22

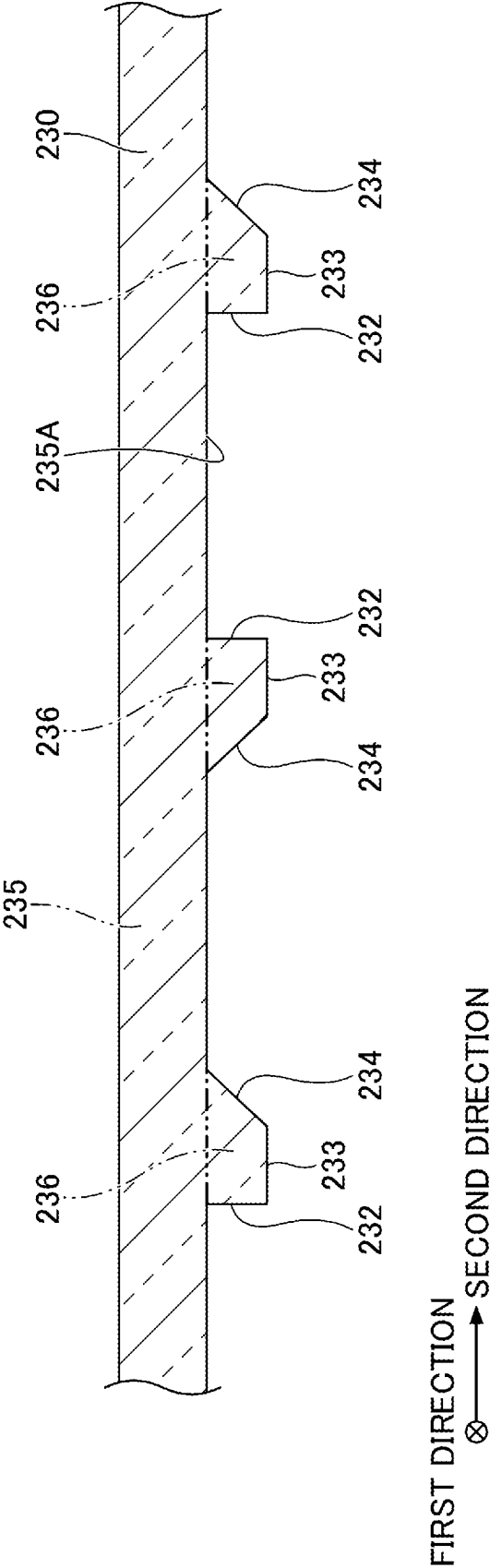


FIG.23

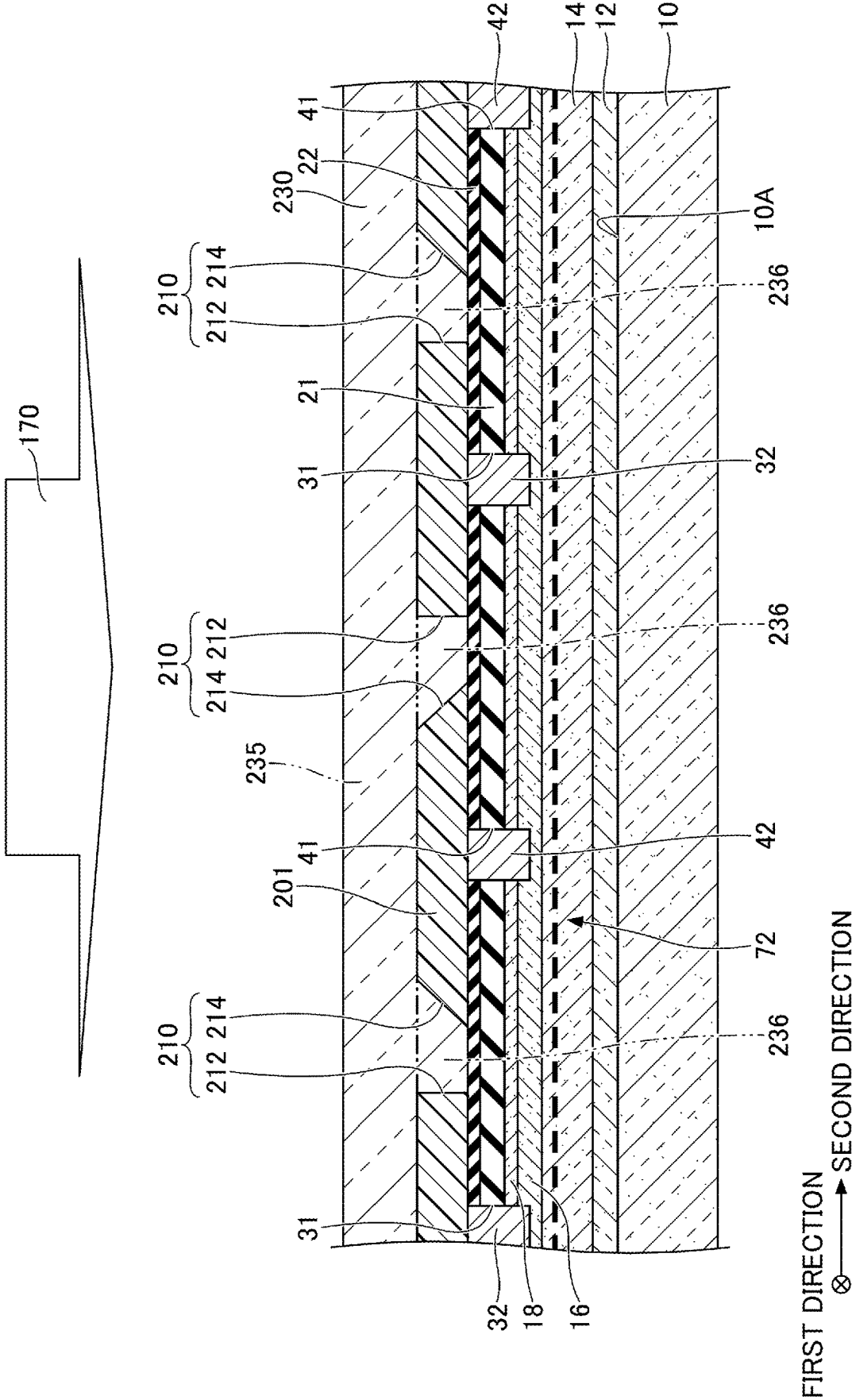


FIG.24

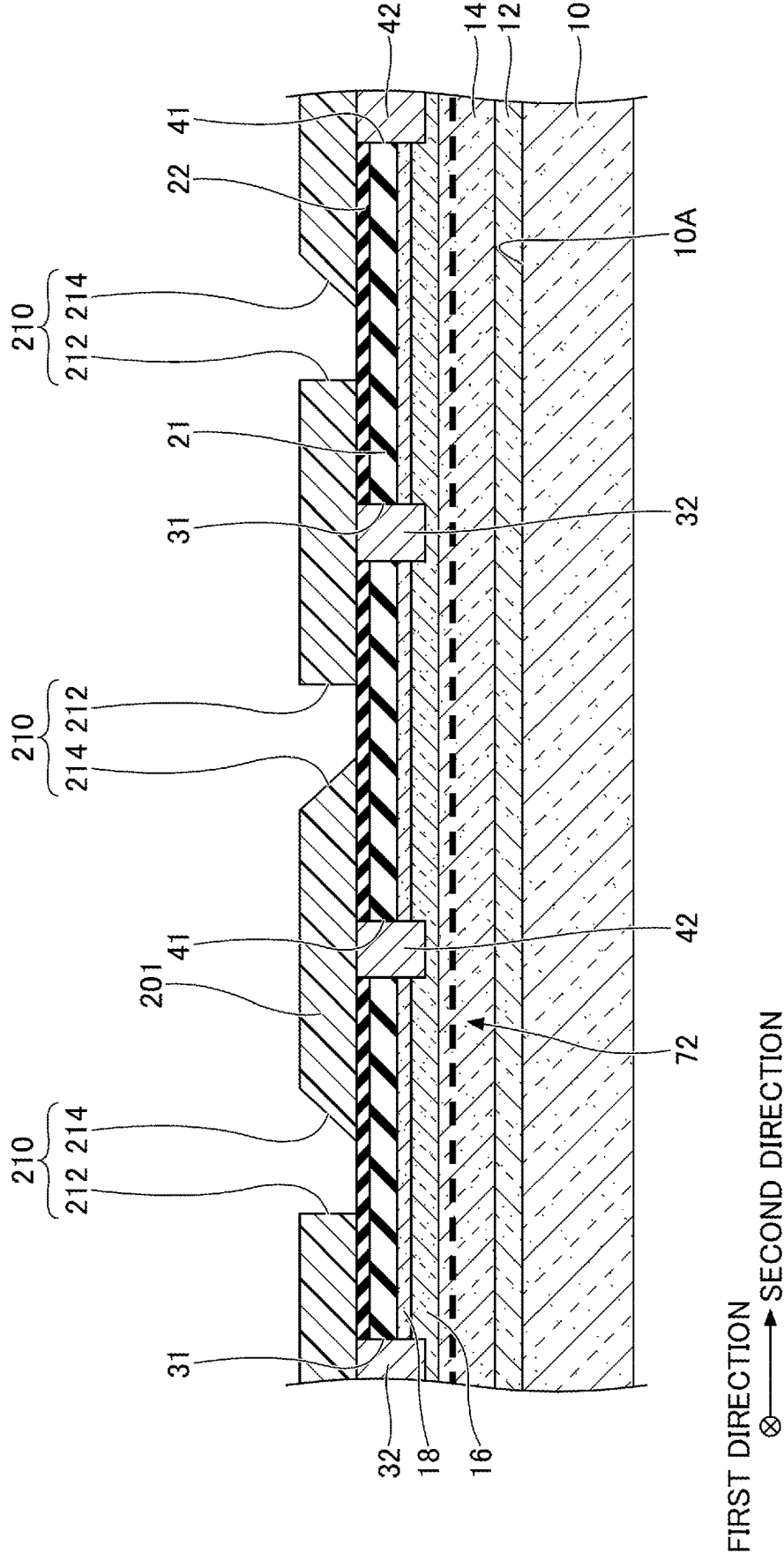


FIG.25

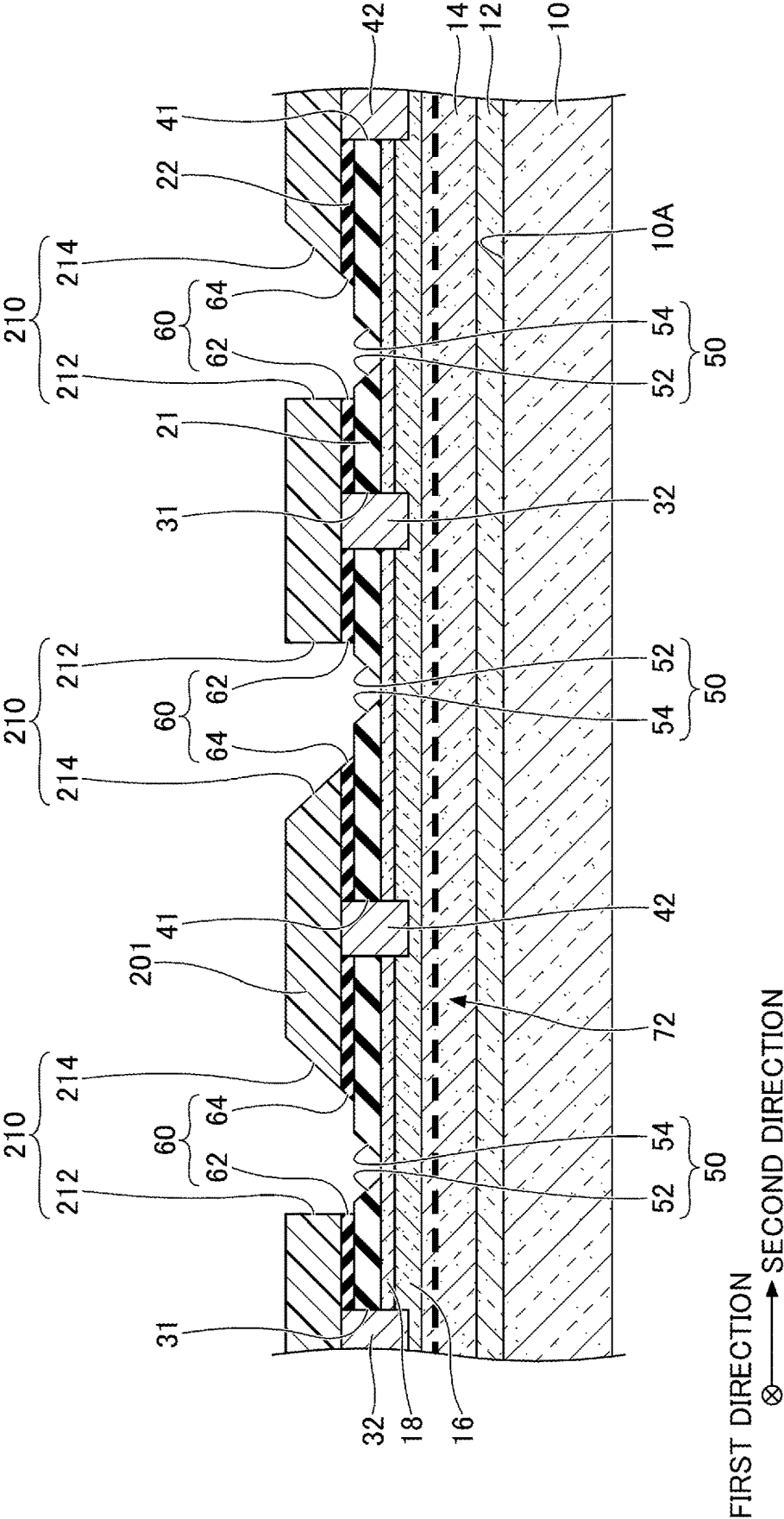


FIG.26

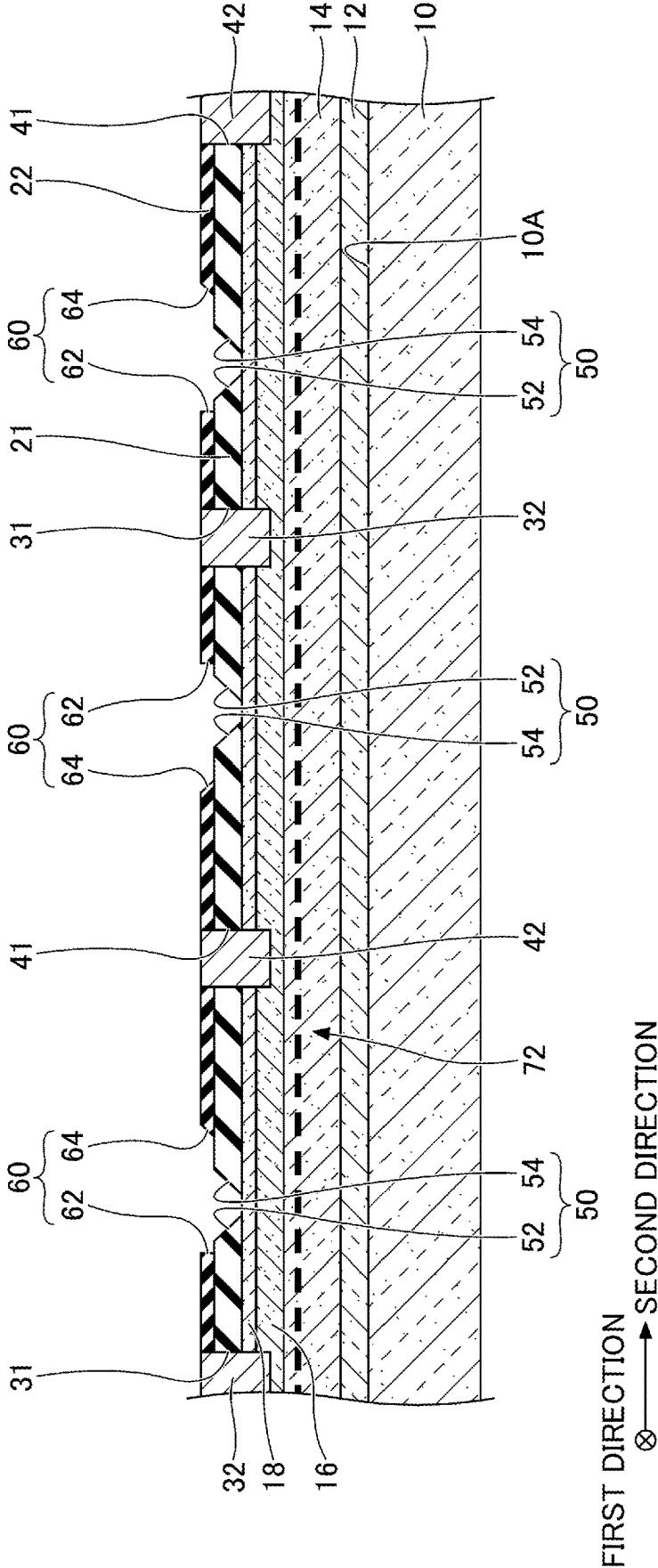


FIG.27

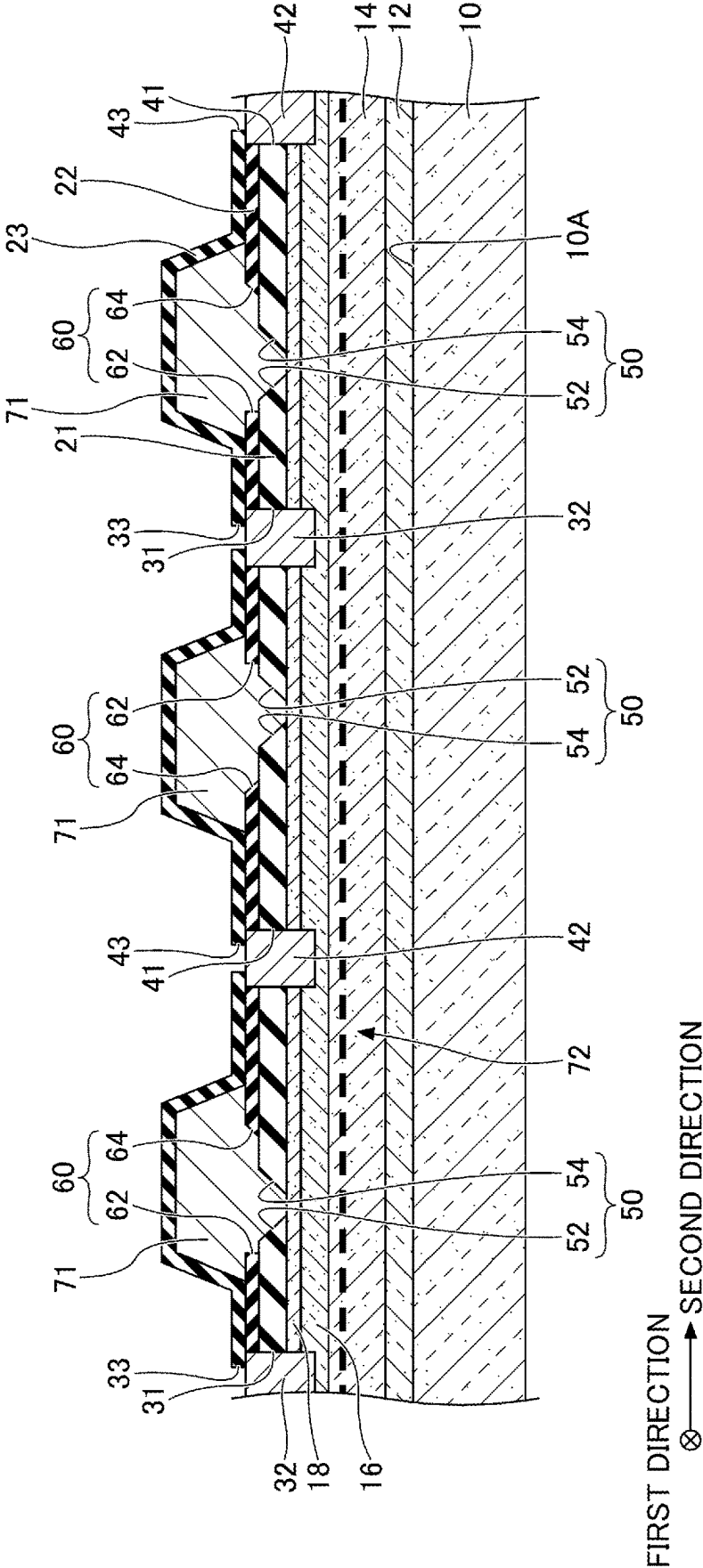


FIG.28

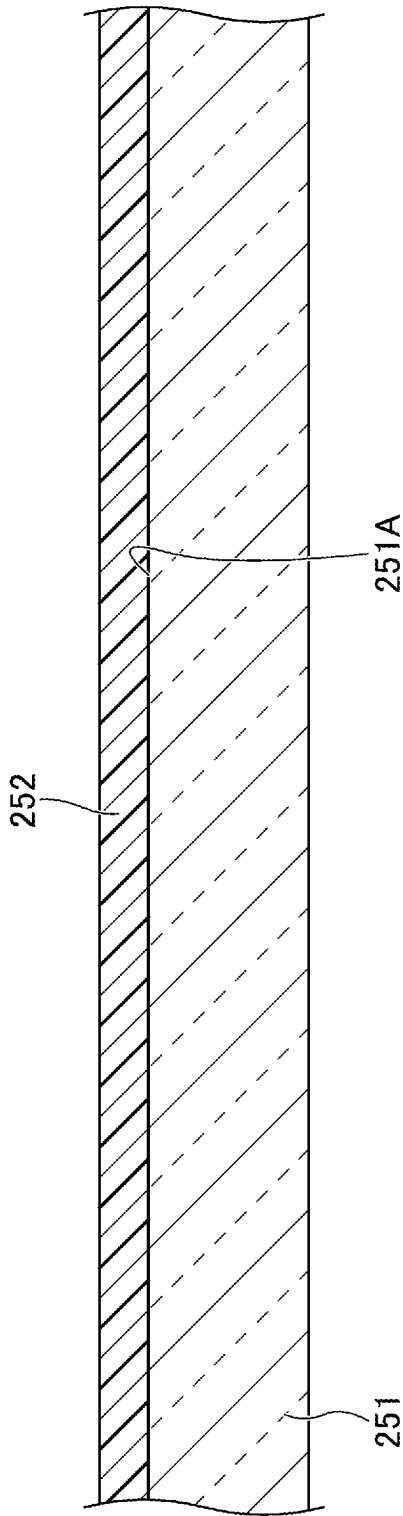


FIG.29

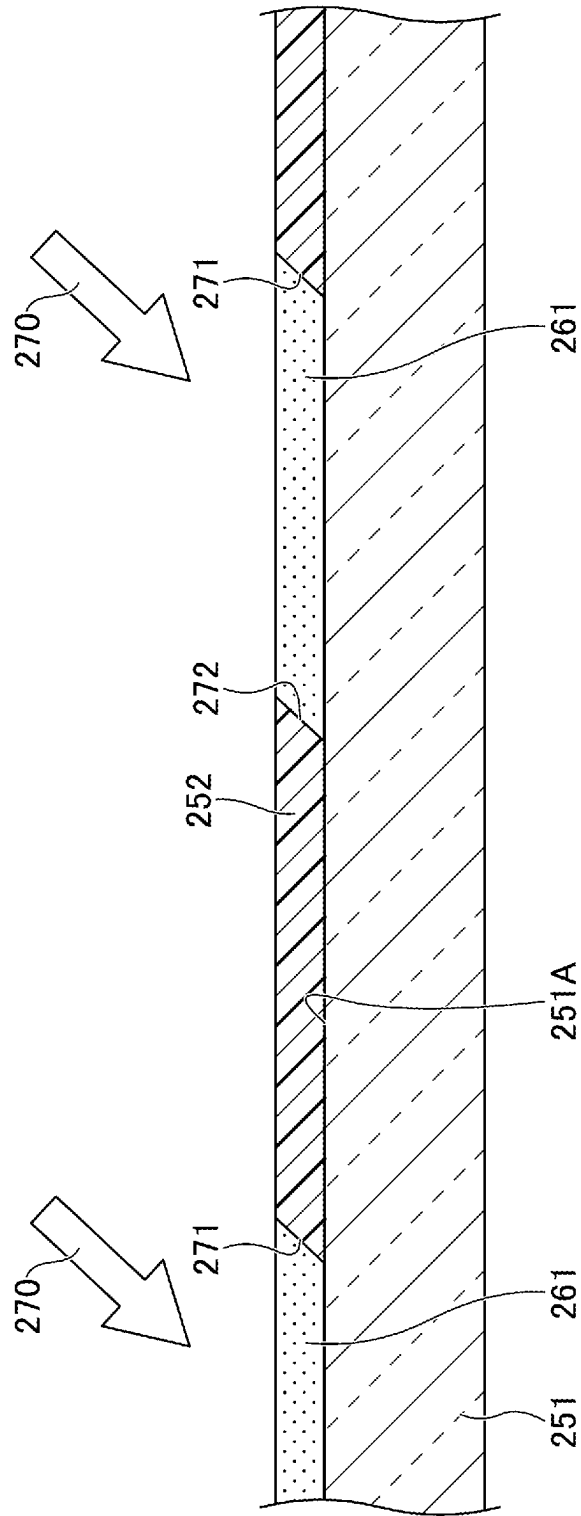


FIG.30

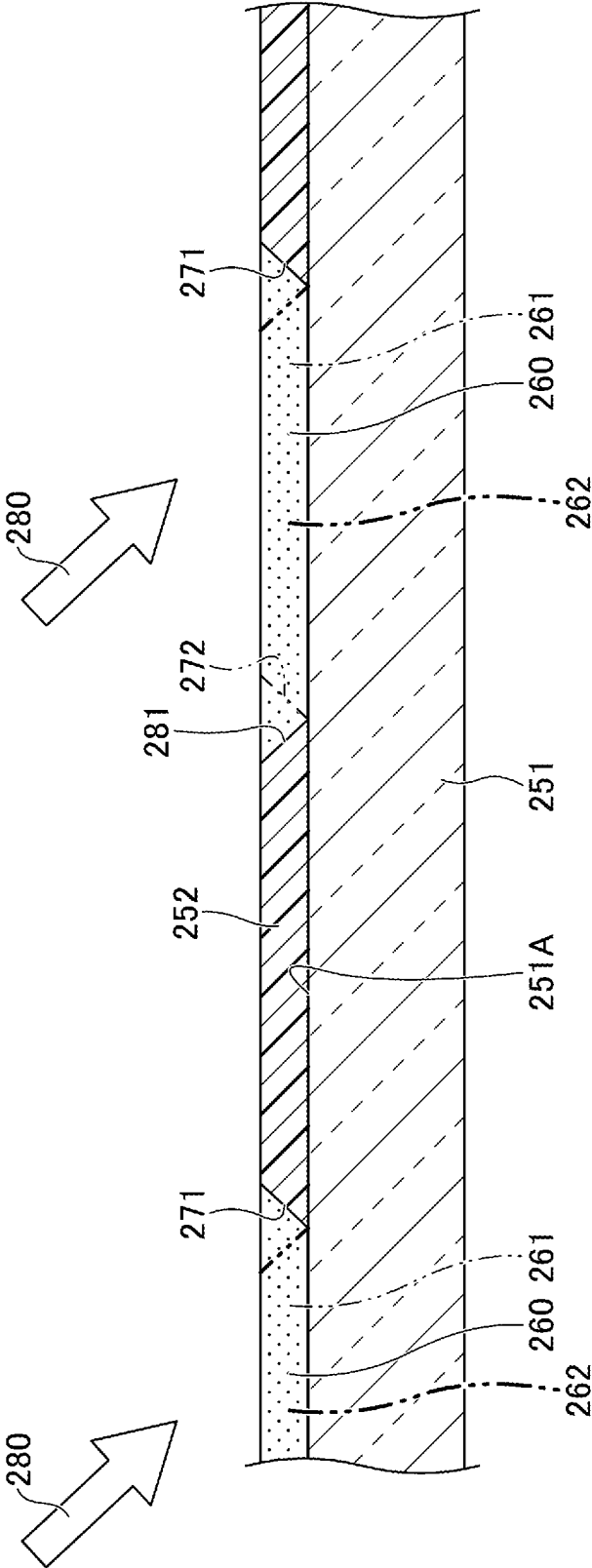


FIG.31

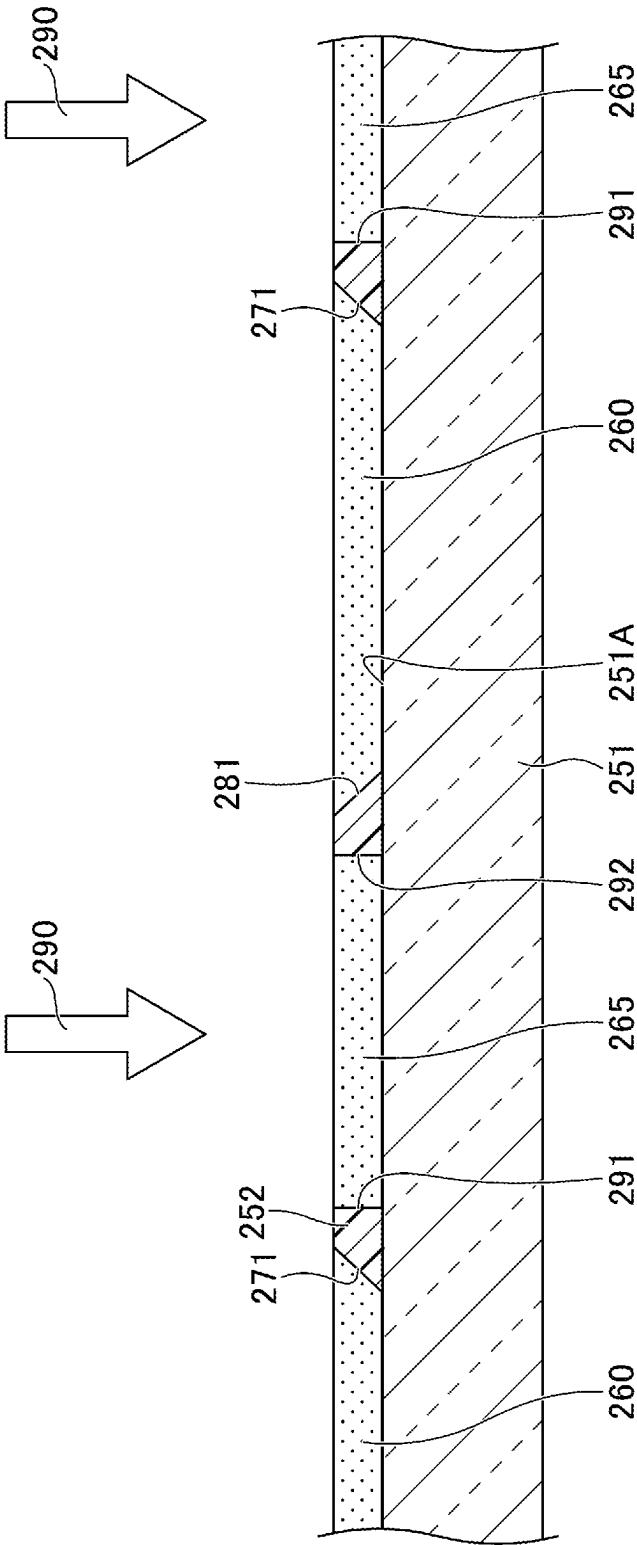


FIG.32

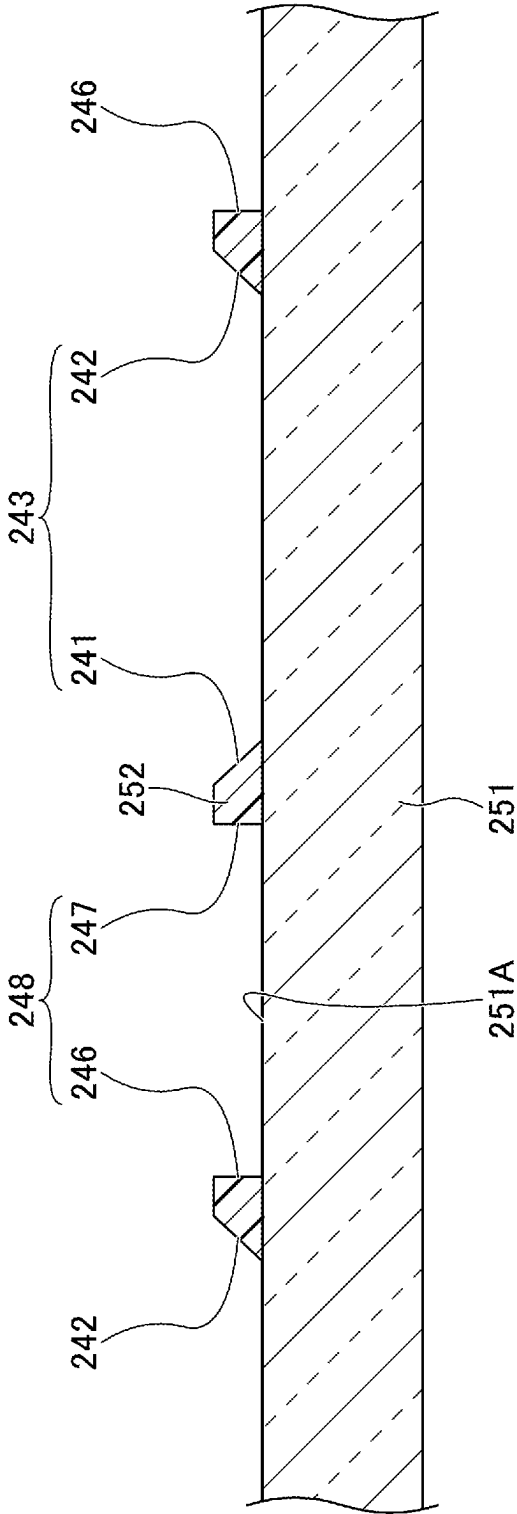


FIG.33

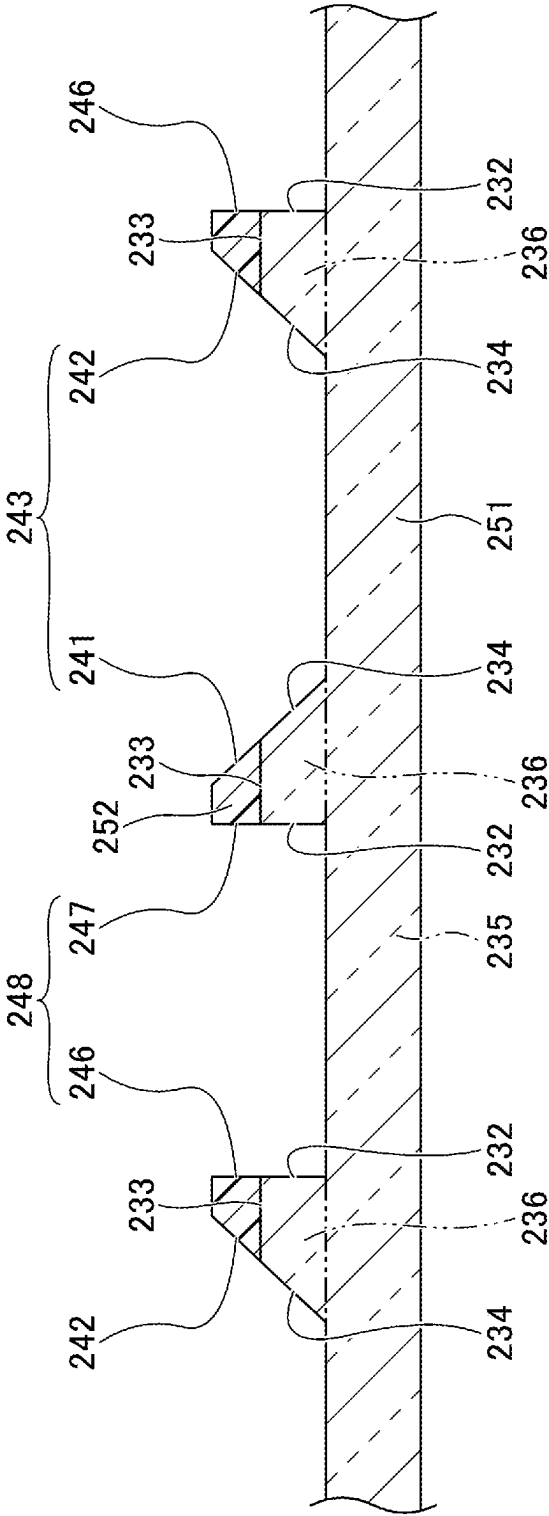


FIG.34

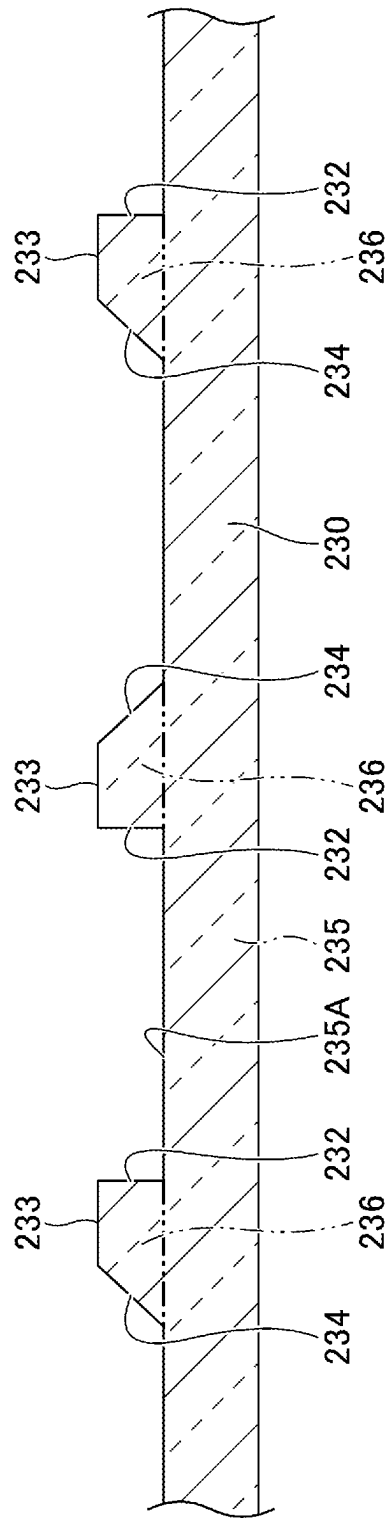


FIG.35

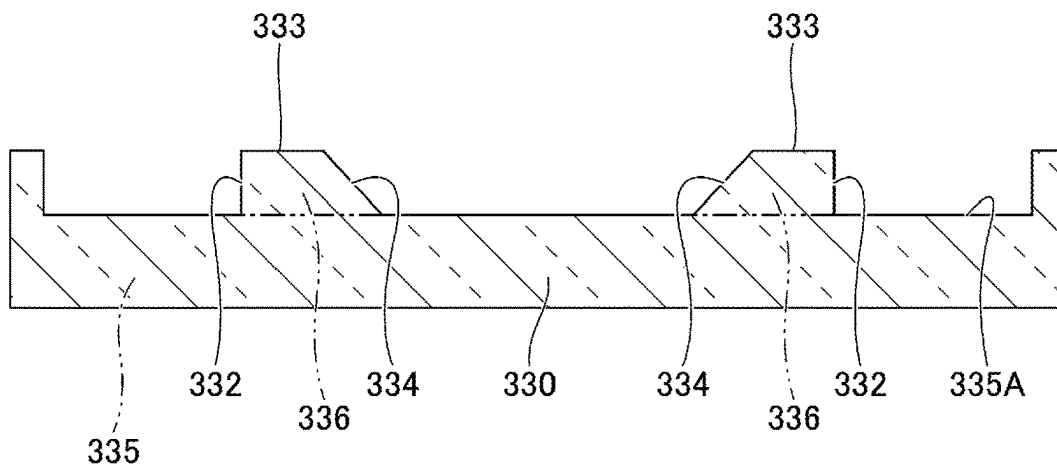


FIG.36

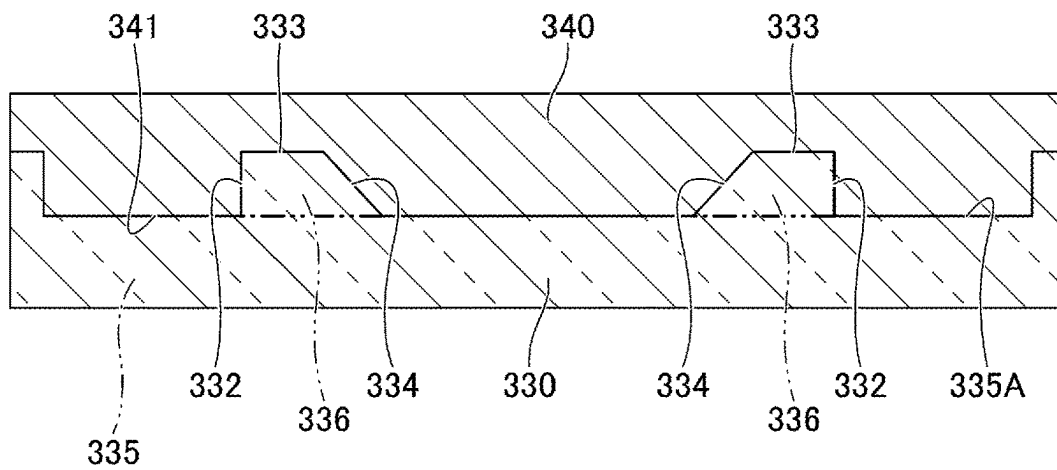


FIG.37

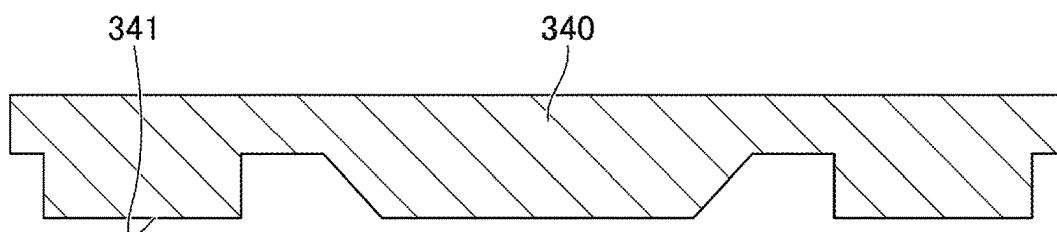


FIG.38

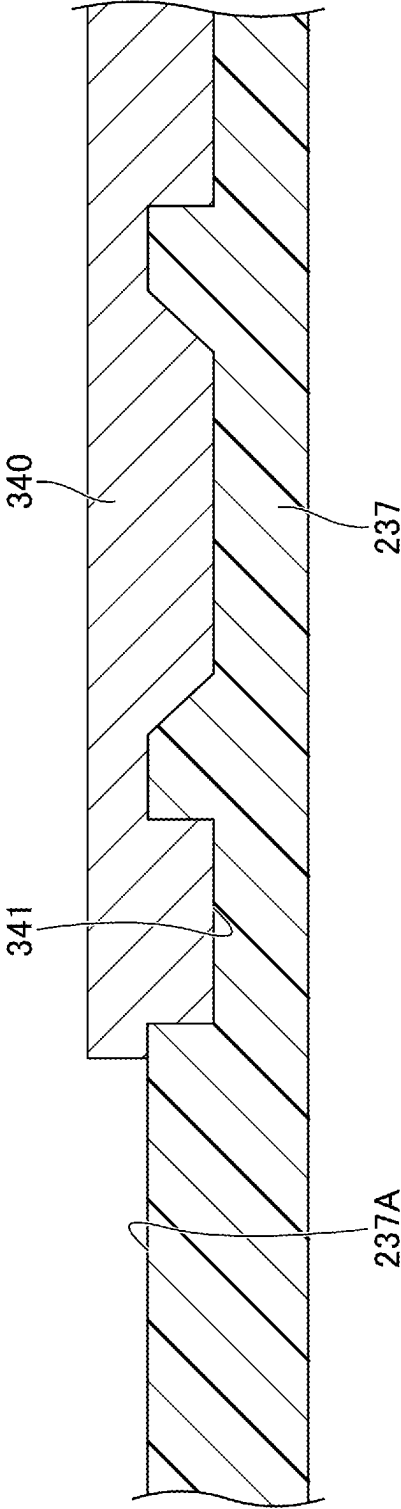
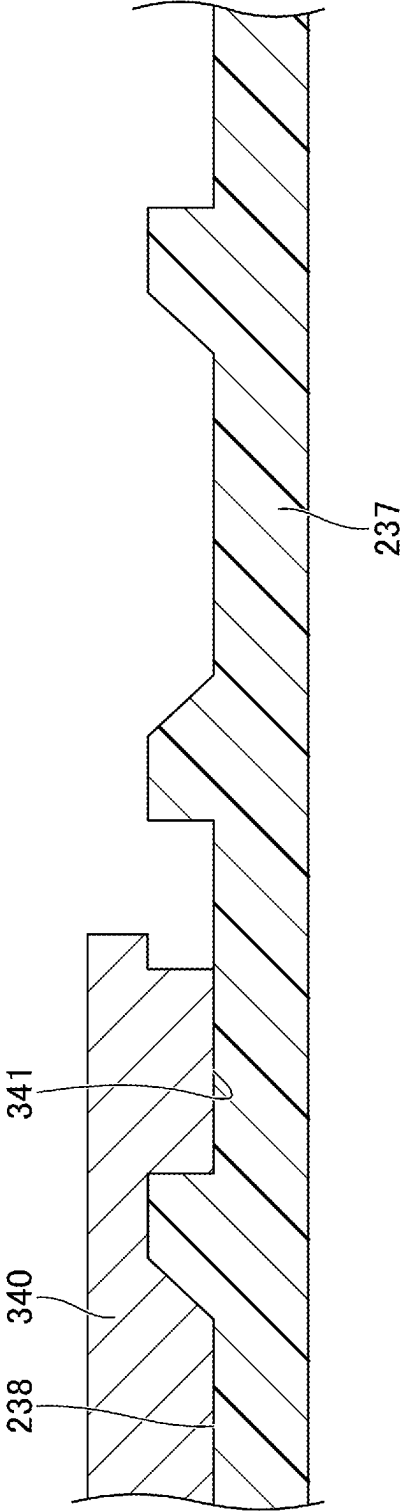


FIG.39



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METHOD FOR MANUFACTURING SEMICONDUCTOR DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is based upon and claims priority to Japanese Patent Application No. 2022-038029, filed on Mar. 11, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to methods for manufacturing semiconductor devices.

BACKGROUND

For a high electron mobility transistor (HEMT), there is a proposed structure in which a portion of a gate electrode extending on an insulating film toward a drain electrode is larger than a portion of the gate electrode extending on the insulating film toward a source electrode, in order to reduce electric field concentration. If the portion of the gate electrode extending on the insulating film toward the source electrode were large, an electrostatic capacitance between the gate electrode and a channel (two-dimensional electron gas) would become large.

Related art include Japanese Laid-Open Patent Publication No. 2013-77621, Japanese Laid-Open Patent Publication No. 2016-157983, Japanese Laid-Open Patent Publication No. 2007-88371, U.S. Pat. Nos. 7,501,669, and 7,812,369, for example.

According to related methods for manufacturing semiconductor devices, it is difficult to simultaneously relax the electric field concentration and reduce an increase in the electrostatic capacitance.

SUMMARY

It is one object of the present disclosure to provide a method for manufacturing a semiconductor device capable of simultaneously relaxing of the electric field concentration and reducing an increase in the electrostatic capacitance.

According to one aspect of the present disclosure, a method for manufacturing a semiconductor device, includes forming a semiconductor layer above a substrate; forming a source electrode and a drain electrode on the semiconductor layer; forming a first insulating film covering a surface of the semiconductor layer, between the source electrode and the drain electrode; forming a second insulating film on the first insulating film; forming a mask on the second insulating film, the mask having an opening between the source electrode and the drain electrode in a plan view viewed in a direction perpendicular to an upper surface of the substrate; forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first insulating film and the second insulating film through the opening; and forming a gate electrode on the first insulating film and the second insulating film, the gate electrode making a Schottky contact with the semiconductor layer through the first gate opening and the second gate opening, wherein the opening includes a first side surface, and a second side surface closer to the drain electrode than the first side surface is to the drain electrode, an angle formed between the first side surface and the upper surface is larger than an angle formed between the second

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side surface and the upper surface, and an etching rate of the second insulating film is higher than an etching rate of the first insulating film during the etching of the forming the first gate opening.

The object and advantages of the embodiments will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and not restrictive of the invention, as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a cross sectional view (part 1) illustrating a method for manufacturing a semiconductor device according to a first embodiment;

FIG. 2 is a cross sectional view (part 2) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 3 is a cross sectional view (part 3) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 4 is a cross sectional view (part 4) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 5 is a cross sectional view (part 5) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 6 is a cross sectional view (part 6) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 7 is a sectional view (part 7) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 8 is a cross sectional view (part 8) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 9 is a cross sectional view (part 9) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 10 is a cross sectional view (part 10) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 11 is a cross sectional view (part 11) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 12 is a cross sectional view (part 12) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 13 is a cross sectional view (part 13) illustrating the method for manufacturing the semiconductor device according to the first embodiment;

FIG. 14 is a diagram generally illustrating a finger gate structure;

FIG. 15 is a cross sectional view (part 1) illustrating the method for manufacturing the semiconductor device according to a second embodiment;

FIG. 16 is a cross sectional view (part 2) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 17 is a cross sectional view (part 3) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 18 is a cross sectional view (part 4) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

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FIG. 19 is a sectional view (part 5) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 20 is a cross sectional view (part 6) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 21 is a sectional view (part 7) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 22 is a cross sectional view (part 8) depicting the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 23 is a cross sectional view (part 9) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 24 is a cross sectional view (part 10) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 25 is a cross sectional view (part 11) depicting the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 26 is a cross sectional view (part 12) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 27 is a cross sectional view (part 13) illustrating the method for manufacturing the semiconductor device according to the second embodiment;

FIG. 28 is a cross sectional view (part 1) illustrating a first example of a mold forming method;

FIG. 29 is a cross sectional view (part 2) illustrating the first example of the mold forming method;

FIG. 30 is a cross sectional view (part 3) illustrating the first example of the mold forming method;

FIG. 31 is a cross sectional view (part 4) illustrating the first example of the mold forming method;

FIG. 32 is a cross sectional view (part 5) illustrating the first example of the mold forming method;

FIG. 33 is a cross sectional view (part 6) illustrating the first example of the mold forming method;

FIG. 34 is a cross sectional view (part 7) illustrating the first example of the mold forming method;

FIG. 35 is a cross sectional view (part 1) illustrating a second example of the mold forming method;

FIG. 36 is a cross sectional view (part 2) illustrating the second example of the mold forming method;

FIG. 37 is a cross sectional view (part 3) illustrating the second example of the mold forming method;

FIG. 38 is a cross sectional view (part 4) illustrating the second example of the mold forming method; and

FIG. 39 is a cross sectional view (part 5) illustrating the second example of the mold forming method.

DETAILED DESCRIPTION

A description will hereinafter be given of embodiments of the present invention with reference to the drawings.

[1] A method for manufacturing a semiconductor device according to one aspect of the present disclosure, includes: forming a semiconductor layer above a substrate; forming a source electrode and a drain electrode on the semiconductor layer; forming a first insulating film covering a surface of the semiconductor layer, between the source electrode and the drain electrode; forming a second insulating film on the first insulating film; forming a mask on the second insulating film, the mask having an opening between the source electrode and the drain electrode in a plan view viewed in a direction perpendicular to an upper surface of the substrate;

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forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first insulating film and the second insulating film through the opening; and forming a gate electrode on the first insulating film and the second insulating film, the gate electrode making a Schottky contact with the semiconductor layer through the first gate opening and the second gate opening, wherein the opening includes a first side surface, and a second side surface closer to the drain electrode than the first side surface is to the drain electrode, an angle formed between the first side surface and the upper surface is larger than an angle formed between the second side surface and the upper surface, and an etching rate of the second insulating film is higher than an etching rate of the first insulating film during the etching of the forming the first gate opening.

In the opening formed in the mask, the angle formed between the first side surface on the side closer to the source electrode and the upper surface of the substrate, is larger than the angle formed between the second side surface on the side closer to the drain electrode and the upper surface of the substrate. For this reason, the opening of the mask can easily widen on the side closer to the drain electrode than on the side closer to the source electrode during etching. Further, during the etching, the etching rate of the second insulating film is higher than the etching rate of the first insulating film. Hence, the second gate opening formed in the second insulating film is more easily affected by the shape of the opening of the mask than the first gate opening formed in the first insulating film. Accordingly, the portion of the first insulating film exposed from the second insulating film can easily widen on the side closer to the drain electrode than on the side closer to the source electrode. Thus, the portion of the gate electrode extending on the first insulating film toward the drain electrode can be made larger than the portion of the gate electrode extending on the first insulating film toward the source electrode. As a result, it is possible to simultaneously relax the electric field concentration and reduce the increase in the electrostatic capacitance.

[2] In [1], a distance between the first gate opening and the second gate opening on a side closer to the drain electrode than the first gate is to the drain electrode opening may be longer than a distance between the first gate opening and the second gate opening on a side closer to the source electrode than the first gate opening is to the source electrode. In this case, it is easy to simultaneously relax of the electric field concentration and reduce of an increase in the electrostatic capacitance.

[3] In [1] or [2], an angle between the first side surface and the upper surface may be in a range greater than or equal to 85° and less than or equal to 90° , and an angle between the second side surface and the upper surface may be in a range greater than or equal to 45° and less than or equal to 60° . In this case, particularly during the etching, the opening of the mask can easily widen on the side closer to the drain electrode than on the side closer to the source electrode.

[4] In any one of [1] to [3], the first insulating film may be a first silicon nitride film having a first refractive index, and the second insulating film may be a second silicon nitride film having a second refractive index lower than the first refractive index. In this case, the etching rate of the second insulating film can easily be made higher than the etching rate of the first insulating film.

[5] In any one of [1] to [4], the forming the mask may include forming a positive photosensitive film on the second insulating film, exposing the photosensitive film to form a

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photosensitive region in a portion of the photosensitive film where the opening is to be formed, and removing the photosensitive region by developing the photosensitive film, and the exposing the photosensitive film may include an exposure from a direction parallel to the first side surface, and exposure from a direction parallel to the second side surface. In this case, it is easy to form the opening with high accuracy.

[6] In any one of [1] to [4], the forming the mask may include forming a positive photosensitive film on the second insulating film, exposing the photosensitive film by a single exposure using a shifted telecentric optical system, to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and removing the photosensitive region by developing the photosensitive film. In this case, the opening can easily be formed with a particularly high accuracy.

[7] In any one of [1] to [4], the forming the mask may include planarizing an upper surface of the second insulating film, forming an uncured film of a resin on the planarized second insulating film, forming a cured film by curing the uncured film while pressing a mold against the uncured film, and removing the mold from the cured film, and the mold may include a base, and a protrusion protruding from the base and having a shape corresponding to a shape of the opening. In this case, the mask can easily be formed with a high accuracy.

[8] In [7], the resin may be an ultraviolet curable resin, and the forming the cured film may include irradiating ultraviolet light on the uncured film through the mold. In this case, the mask can easily be formed with a high accuracy.

[9] In [7] or [8], the source electrode and the drain electrode may extend in a first direction parallel to the upper surface, and be alternately formed at multiple locations in a second direction parallel to the upper surface and perpendicular to the first direction, and the gate electrode may be formed at each location between the source electrode and the drain electrode that are adjacent to each other in the second direction. In this case, a transistor having a finger gate structure can easily be formed.

[10] In any one of [1] to [9], an etching rate of the second insulating film may be four times or more higher than an etching rate of the first insulating film during the etching of the forming the first gate opening. In this case, the portion of the first insulating film exposed from the second insulating film can easily be made larger on the side closer to the drain electrode than on the side closer to the source electrode.

Details of Embodiments of Present Disclosure

Hereinafter, embodiments of the present disclosure will be described in detail, but the present disclosure is not limited to the described embodiments. In the present specification and drawings, constituent elements having substantially the same functional configuration are denoted by the same reference numerals, a redundant description thereof may be omitted.

First Embodiment

A first embodiment will be described. The first embodiment relates to a method for manufacturing a semiconductor device including a GaN-HEMT having a nitride semiconductor as a main constituent material thereof. FIG. 1 through FIG. 13 are cross sectional views illustrating the method for manufacturing the semiconductor device according to the first embodiment.

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First, as illustrated in FIG. 1, a buffer layer 12, an electron transit layer (or channel layer) 14, an electron supply layer (or barrier layer) 16, and a capping layer 18 are successively formed on a substrate 10. The substrate 10 is a silicon carbide (SiC) substrate having an upper surface that is (0001)-oriented, for example. The buffer layer 12 is an aluminum nitride (AlN) layer having a thickness in a range greater than or equal to 5 nm and less than or equal to 100 nm, for example. The electron transit layer 14 is an undoped gallium nitride (GaN) layer having a thickness of approximately 1000 nm, for example. The electron supply layer 16 is an n-type aluminum gallium nitride (AlGaIn) layer having a thickness of approximately 20 nm, for example. The capping layer 18 is an n-type GaN layer having a thickness of approximately 5 nm, for example. An n-type impurity used in this embodiment is silicon (Si) or germanium (Ge), for example. A laminating direction of the buffer layer 12, the electron transit layer 14, the electron supply layer 16, and the capping layer 18 is the [0001]-direction, for example. The buffer layer 12, the electron transit layer 14, the electron supply layer 16, and the capping layer 18 are formed by metal organic chemical vapor deposition (MOCVD), for example. A two-dimensional electron gas (2DEG) 72 is present near an upper surface of the electron transit layer 14. A laminated structure formed by the buffer layer 12, the electron transit layer 14, the electron supply layer 16, and the capping layer 18, is an example of a semiconductor layer.

Next, a first insulating film 21 is formed on the capping layer 18. The first insulating film 21 is a silicon nitride (SiN) film, for example. The first insulating film 21 has a thickness in a range greater than or equal to 10 nm and less than or equal to 20 nm, for example. The silicon nitride film (an example of a first silicon nitride film) used for the first insulating film 21 is preferably has a composition represented by a formula Si_2N_3 . A refractive index of the Si_2N_3 is 2.0. The first insulating film 21 can be formed by chemical vapor deposition (CVD), for example.

Next, as illustrated in FIG. 2, a resist mask 80 is formed on the first insulating film 21. The resist mask 80 has an opening 81 that exposes a portion of the first insulating film 21, and an opening 82 that exposes another portion of the first insulating film 21.

Next, as illustrated in FIG. 3, openings 31 and 41 are formed in the first insulating film 21, the capping layer 18, and the electron supply layer 16 by reactive ion etching (RIE). The opening 31 is communicates to the opening 81, and the opening 41 communicates to the opening 82. A reactive gas including fluorine (F) may be used to etch the first insulating film 21. A reactive gas including chlorine (Cl) may be used to etch the capping layer 18 and the electron supply layer 16. The openings 31 and 41 are formed to a depth such that a portion of the electron supply layer 16 remains, for example.

Next, as illustrated in FIG. 4, a source electrode 32 is formed in the opening 31, and a drain electrode 42 is formed in the opening 41. When forming the source electrode 32 and the drain electrode 42, a metal film is formed by vacuum deposition in a state where the resist mask 80 remains, and the resist mask 80 is removed after forming the metal film. Although the metal film is formed not only inside the openings 31 and 41 but also on the resist mask 80, the metal film on the resist mask 80 is removed when the resist mask 80 is removed. In other words, lift-off of the metal film together with the resist mask 80 is performed. The resist mask 80 can be removed using an organic solvent, for example. An alloying heat treatment is performed after the resist mask 80 is removed. The source electrode 32 and the

drain electrode 42 include a Ta film and an Al film, for example. The source electrode 32 and the drain electrode 42 make ohmic contact with the 2DEG 72.

Next, as illustrated in FIG. 5, a second insulating film 22 is formed on the first insulating film 21, the source electrode 32, and the drain electrode 42. The second insulating film 22 is a silicon nitride (SiN) film, for example. The second insulating film 22 may be thicker than the first insulating film 21. The second insulating film 22 has a thickness in a range greater than or equal to 30 nm and less than or equal to 40 nm, for example. A density of the second insulating film 22 is preferably lower than a density of the first insulating film 21. In addition, a refractive index (an example of a second refractive index) of the second insulating film 22 is preferably lower than the refractive index (an example of a first refractive index) of the first insulating film 21. The refractive index of the silicon nitride film (an example of a second silicon nitride film) used for the second insulating film 22 is 1.85, for example. The second insulating film 22 can be formed by CVD, for example.

Next, as illustrated in FIG. 6, a positive photoresist film 101 is formed on the second insulating film 22. The photoresist film 101 is an example of a photosensitive film.

Next, as illustrated in FIG. 7, a photosensitive region 121 is formed, by exposure, in a portion of the photoresist film 101 between the source electrode 32 and the drain electrode 42. The exposure is performed by reduction exposure, for example. When performing the exposure, light 150 is irradiated from a direction inclined from a direction perpendicular to an upper surface 10A of the substrate 10 toward the drain electrode 42. The photosensitive region 121 has side surfaces 151 and 152 that are parallel to each other in a cross sectional view thereof. The side surface 151 is closer to the source electrode 32 than the side surface 152 is to the source electrode 32. An angle formed by the upper surface 10A and the side surfaces 151 and 152 may be in a range greater than or equal to 45° and less than or equal to 60°, for example. The photosensitive region 121 has a cross section having a parallelogram shape.

Next, as illustrated in FIG. 8, a photosensitive region 122 partially overlapping the photosensitive region 121 is formed, by exposure, in a portion of the photoresist film 101 between the source electrode 32 and the drain electrode 42. The exposure is performed by reduction exposure, for example. When performing the exposure, light 160 is irradiated from the direction perpendicular to the upper surface 10A of the substrate 10. The photosensitive region 122 is formed such that the side surface 152 remains, but the side surface 151 is incorporated into the photosensitive region 122. The photosensitive region 122 has a side surface 161 located at a position closer to the source electrode 32, than the position where the side surface 151 existed is to the source electrode 32. Although the upper surface 10A and the side surface 161 are perpendicular to each other, the side surface 161 may be slightly inclined from the direction perpendicular to the upper surface 10A of the substrate 10 toward the source electrode 32 or the drain electrode 42. An angle formed by the upper surface 10A and the side surface 161 is in a range greater than or equal to 85° and less than or equal to 90°, for example. The photosensitive region 122 has a cross section having a rectangular shape.

A photosensitive region 120 is formed from the photosensitive regions 121 and 122 partially overlapping each other. The photosensitive region 120 has the side surfaces 152 and 161. The photosensitive region 120 has a cross section having a trapezoidal shape with two adjacent apexes forming an angle of 90°, where one of the two adjacent

apexes is formed by the side surface 161 and an upper surface of the photosensitive region 122, and the other of the two adjacent apexes is formed by the side surface 161 and a lower surface of the photosensitive region 122.

Next, as illustrated in FIG. 9, the photosensitive region 120 is removed by developing the photoresist film 101. As a result, an opening 110 having side surfaces 112 and 114 is formed in the photoresist film 101. The side surface 112 is formed at the position of the side surface 161 of the photosensitive region 120, and the side surface 114 is formed at the position of the side surface 152 of the photosensitive region 120. Accordingly, the side surface 112 is closer to the source electrode 32 than the side surface 114 is to the source electrode 32. In the opening 110, an angle formed by the side surface 112 on the side of the source electrode 32 and the upper surface 10A of the substrate 10 is larger than an angle formed by the side surface 114 on the side of the drain electrode 42 and the upper surface 10A of the substrate 10. The angle formed by the upper surface 10A and the side surface 112 may be in a range greater than or equal to 85° and less than or equal to 90° for example, and the angle formed by the upper surface 10A and the side surface 114 may be in a range greater than or equal to 45° and less than or equal to 60°, for example. An alkaline developer is used to develop the photoresist film 101, for example. The photoresist film 101 formed with the opening 110 is an example of a mask.

Next, as illustrated in FIG. 10, the first gate opening 50 is formed in the first insulating film 21, and the second gate opening 60 is formed in the second insulating film 22, by etching the first insulating film 21 and the second insulating film 22 through the opening 110. RIE is used for this etching, for example. The first gate opening 50 has side surfaces 52 and 54 in the cross sectional view. The side surface 52 is located at a position closer to the source electrode 32 than the side surface 54 is to the source electrode 32. The second gate opening 60 has side surfaces 62 and 64 in the cross sectional view. The side surface 62 is located at a position closer to the source electrode 32 than the side surface 64 is to the source electrode 32. A reactive gas including fluorine (F) may be used to etch the first insulating film 21 and the second insulating film 22. This etching is performed under a condition that the etching rate of the second insulating film 22 becomes higher than the etching rate of the first insulating film 21. For example, the etching rate of the second insulating film 22 is set to approximately four times the etching rate of the first insulating film 21. For example, the etching rate of the first insulating film 21 is in a range greater than or equal to 1 nm/minute and less than or equal to 3 nm/minute, and the etching rate of the second insulating film 22 is in a range greater than or equal to 4 nm/minute and less than or equal to 12 nm/minute.

During this etching, the opening 110 of the photoresist film 101 widens while maintaining the shape of the side surfaces 112 and 114. In this state, the side surface 112 is perpendicular to the upper surface 10A, but the side surface 114 is inclined from the side surface 112. For this reason, the side surface 114 is more easily etched than the side surface 112, and a moving amount of the side surface 114 is larger than a moving amount of the side surface 112. In addition, the second gate opening 60 of the second insulating film 22 is formed following the shape of the opening 110. Accordingly, the side surface 62 of the second gate opening 60 becomes continuous with the side surface 112, and the side surface 64 of the second gate opening 60 becomes continuous with the side surface 114. The angle formed by the upper surface 10A and the side surface 62 may be in a range greater

than or equal to 85° and less than or equal to 90° , for example, and the angle formed by the upper surface 10A and the side surface 64 may be in a range greater than or equal to 45° and less than or equal to 60° , for example.

On the other hand, because the first insulating film 21 is etched at a rate that is approximately 1/4 that of the second insulating film 22, even when the opening 110 is widened, the first gate opening 50 is hardly affected by the widening of the opening 110. Accordingly, the first gate opening 50 is mainly formed below the portion exposed inside the opening 110 at the start of etching the second insulating film 22. Hence, the side surface 52 of first gate opening 50 becomes discontinuous with the side surface 62, and side surface 54 of first gate opening 50 becomes discontinuous with the side surface 64. The side surfaces 52 and 54 may be a surface perpendicular to the upper surface 10A, or may be a surface inclined from the surface perpendicular to the upper surface 10A.

Because the first gate opening 50 and the second gate opening 60 are formed by such a mechanism, a distance L_s between the side surface 52 of the first gate opening 50 and the side surface 62 of the second gate opening 60 becomes smaller than a distance L_d between the side surface 54 of the first gate opening 50 and the side surface 64 of the second gate opening 60.

After forming the first gate opening 50 and the second gate opening 60, the photoresist film 101 is removed as illustrated in FIG. 11. The photoresist film 101 can be removed by using an organic solvent, oxygen plasma, or the like, for example.

Next, as illustrated in FIG. 12, a gate electrode 71 is formed on the first insulating film 21 and the second insulating film 22. The gate electrode 71 can be formed by vapor deposition and lift-off, for example. The gate electrode 71 includes a Ni film and a Au film, for example. The gate electrode 71 makes a Schottky contact with the capping layer 18, through the first gate opening 50 and the second gate opening 60.

Next, as illustrated in FIG. 13, a third insulating film 23 covering the gate electrode 71, the source electrode 32, and the drain electrode 42, is formed. The third insulating film 23 is a silicon nitride (SiN) film, for example. The silicon nitride film can be formed by CVD, for example. Next, an opening 33 that exposes a portion of the source electrode 32, and an opening 43 that exposes a portion of the drain electrode 42, are formed in the third insulating film 23 and the second insulating film 22. The openings 33 and 43 are formed by RIE using a resist mask (not illustrated), for example.

Thereafter, interconnects or the like are formed, as required. Hence, the semiconductor device 100 including the GaN-HEMT can be manufactured in the above described manner.

According to the first embodiment, in the opening 110 formed in the photoresist film 101, the angle formed by the side surface 112 on the side of the source electrode 32 and the upper surface 10A of the substrate 10 is larger than the angle formed by the side surface 114 on the side of the drain electrode 42 and the upper surface 10A of the substrate 10. For this reason, as described above, the opening 110 easily widens on the side of the drain electrode 42 during the etching than on the side of the source electrode 32. In addition, during the etching, the etching rate of the second insulating film 22 is higher than the etching rate of the first insulating film 21. Hence, the second gate opening 60 is more easily affected by the shape of the opening 110 than the first gate opening 50. Thus, the portion of the first insulating

film 21 exposed from the second insulating film 22 easily widens on the side of the drain electrode 42 than on the side of the source electrode 32. For this reason, the portion of the gate electrode 71 widening on the first insulating film 21 toward the drain electrode 42 can be made larger than the portion of the gate electrode 71 widening on the first insulating film 21 toward the source electrode 32. As a result, it is possible to simultaneously relax the electric field concentration and reduce the increase in electrostatic capacitance.

Moreover, when forming the opening 110, two exposures with different light irradiating directions are performed, and thus, the opening 110 can be easily be formed with a high accuracy. The photosensitive region 120 may be formed by a single exposure using a shifted telecentric optical system, instead of performing the two exposures. It is also possible in this case to easily form the opening 110 with a high accuracy.

When the angle formed by the upper surface 10A and the side surface 112 is in a range greater than or equal to 85° and less than or equal to 90° , and the angle formed by the upper surface 10A and the side surface 114 is in a range greater than or equal to 45° and less than or equal to 60° , the opening 110 in particular easily widens on the side of the drain electrode 42 than on the side of the source electrode 32 during etching. The angle formed by the upper surface 10A and the side surface 112 is more preferably in a range greater than or equal to 87° and less than or equal to 90° , and the angle formed by the upper surface 10A and the side surface 114 is more preferably in a range greater than or equal to 45° and less than or equal to 55° .

In the case where the first insulating film 21 is a first silicon nitride film having a first refractive index (for example, 2.00), and the second insulating film 22 is a second silicon nitride film having a second refractive index (for example, 1.85) lower than the first refractive index, the etching rate of the second insulating film 22 can easily be made higher than the etching rate of the first insulating film 21. The refractive index of the silicon nitride film can be measured using a spectroscopic ellipsometer or a spectroscopic reflectometry film thickness measuring device. Further, the silicon nitride film has wavelength dispersion. The refractive index of the silicon nitride film in the present disclosure is defined as the refractive index at a wavelength of 632.8 nm (HeNe laser).

In the case where the etching rate of the second insulating film 22 is four times or more higher than the etching rate of the first insulating film 21, the portion of the first insulating film 21 exposed from the second insulating film 22 can easily be widened on the side of the drain electrode 42 than on the side of the source electrode 32. More preferably, the etching rate of the second insulating film 22 is five times or more higher than the etching rate of the first insulating film 21.

Second Embodiment

Next, a second embodiment will be described. The second embodiment relates to a method for manufacturing a semiconductor device including a GaN-HEMT having a nitride semiconductor as a main constituent material thereof. The second embodiment is particularly suitable for a GaN-HEMT having a finger gate structure. FIG. 14 is a diagram generally illustrating the finger gate structure. FIG. 15 through FIG. 27 are cross sectional views illustrating the method for manufacturing the semiconductor device according to the second embodiment.

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In the finger gate structure, as illustrated in FIG. 14, the gate electrode 71, the source electrode 32, and the drain electrode 42 extend in a first direction parallel to the upper surface of the substrate. In addition, a plurality of source electrodes 32 and drain electrodes 42 are alternately formed in a second direction that is parallel to the upper surface of the substrate and is perpendicular to the first direction. One gate electrode 71 is formed between each pair of the source electrode 32 and the drain electrode 42 that are adjacent to each other in the second direction.

First, in the second embodiment, the buffer layer 12, the electron transit layer 14, the electron supply layer 16, and the capping layer 18 are successively formed on the substrate 10, as illustrated in FIG. 15, similar to the first embodiment. Next, the first insulating film 21 is formed on the capping layer 18.

Next, as illustrated in FIG. 16, a resist mask 80 is formed on the first insulating film 21. The resist mask 80 has a plurality of openings 81, and a plurality of openings 82. The openings 81 and the openings 82 are alternately arranged in the second direction.

Next, as illustrated in FIG. 17, the openings 31 and 41 are formed in the first insulating film 21, the capping layer 18, and the electron supply layer 16 by RIE, similar to the first embodiment. In this state, the openings 31 and 41 are formed to extend in the first direction parallel to the upper surface 10A of the substrate 10. In addition, the plurality of openings 31 and 41 are alternately formed in the second direction that is parallel to the upper surface 10A and is perpendicular to the first direction.

Next, as illustrated in FIG. 18, the source electrode 32 is formed inside the opening 31, and the drain electrode 42 is formed inside the opening 41, similar to the first embodiment. In this state, the source electrode 32 and the drain electrode 42 are formed to extend in the first direction. In addition, the plurality of source electrodes 32 and the plurality of drain electrodes 42 are alternately formed in the second direction.

Next, as illustrated in FIG. 19, the second insulating film 22 is formed on the first insulating film 21, the source electrode 32, and the drain electrode 42, similar to the first embodiment.

Next, as illustrated in FIG. 20, an upper surface of the second insulating film 22 is planarized. The upper surface of the second insulating film 22, an upper surface of the source electrode 32, and an upper surface of the drain electrode 42 may coincide with one another. For example, the second insulating film 22 is polished until the upper surface of the source electrode 32 and the upper surface of the drain electrode 42 become exposed. For example, chemical mechanical polishing (CMP) may be performed as the polishing of the second insulating film 22. The source electrode 32 and the drain electrode 42 may also be polished.

Next, as illustrated in FIG. 21, an uncured film 201A of an ultraviolet (UV) curable resin is formed on the second insulating film 22.

Further, as illustrated in FIG. 22, a mold 230 that transmits ultraviolet light is prepared. The mold 230 includes a flat base 235, and a plurality of protrusions 236 protruding from the base 235. The protrusions 236 protrude from one surface 235A of the base 235, that is, from a lower surface of the base 235, for example. The protrusions 236 extend in the first direction, and are arranged side by side in the second direction. The protrusions 236 are provided to form openings corresponding to the openings 110 of the first embodiment, in a mask to be formed from the uncured film 201A. The protrusion 236 has a surface 233 parallel to the surface

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235A, a side surface 232, and a side surface 234. The side surface 232 is perpendicular to the surfaces 233 and 235A, and joins the surface 233 and the surface 235A. The side surface 234 is inclined from the side surface 232, and joins the surface 233 and the surface 235A. The side surfaces 232 and 234 are alternately arranged in the second direction for each of the protrusions 236. There are two kinds of inclination directions of the side surface 234. The mold 230 is a quartz mold, for example. A method for forming the mold 230 will be described later.

After forming the uncured film 201A, the mold 230 is pressed against the uncured film 201A, as illustrated in FIG. 23. Then, ultraviolet light 170 is irradiated on the uncured film 201A through the mold 230. As a result, a cured film 201 of the UV curable resin is formed. The cured film 201 of the UV curable resin has an opening 210 with side surfaces 212 and 214. The side surface 212 is formed at the position of the side surface 232 of the protrusion 236, and the side surface 214 is formed at the position of the side surface 234 of the protrusion 236. The side surface 212 is located at a position closer to the source electrode 32 than the side surface 214 is to the source electrode 32. In the opening 210, an angle formed by the side surface 212 on the side of the source electrode 32 and the upper surface 10A of the substrate 10 is larger than an angle formed by the side surface 214 on the side of the drain electrode 42 and the upper surface 10A of the substrate 10. The angle formed by the upper surface 10A and the side surface 212 may be in a range greater than or equal to 85° and less than or equal to 90°, for example, and the angle formed by the upper surface 10A and the side surface 214 may be in a range greater than or equal to 45° and less than or equal to 60°, for example.

Next, as illustrated in FIG. 24, the mold 230 is removed. The cured film 201 of the UV curable resin, after the mold 230 is removed, is an example of the mask. A residual film treatment (or removal) is preferably performed after removing the mold 230. The protrusions 236 and the second insulating film 22 may inevitably not make close contact with one another, thereby causing a residual UV curable resin to remain at a bottom of the opening 210. But even if the residual UV curable resin is present, the residual UV curable resin can be removed by the residual film treatment. The residual film treatment irradiates oxygen plasma, for example.

Next, as illustrated in FIG. 25, the second gate opening 60 is formed in the second insulating film 22, and the first gate opening 50 is formed in the first insulating film 21 by RIE, similar to the first embodiment. The distance between the side surface 52 of the first gate opening 50 and the side surface 62 of the second gate opening 60 becomes smaller than the distance between the side surface 54 of the first gate opening 50 and the side surface 64 of the second gate opening 60, similar to the first embodiment.

Next, as illustrated in FIG. 26, the cured film 201 of the UV curable resin is removed. The cured film 201 of the UV curable resin can be removed using an organic solvent, oxygen plasma, or the like, for example.

Next, as illustrated in FIG. 27, the gate electrode 71 is formed on the first insulating film 21 and the second insulating film 22, the third insulating film 23 is formed to cover the gate electrode 71, the source electrode 32, and the drain electrode 42, and the openings 33 and 43 are formed in the third insulating film 23, similar to the first embodiment.

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Thereafter, interconnections or the like are formed, as required. Hence, the semiconductor device **200** including the GaN-HEMT can be manufactured in the above described manner.

According to the second embodiment, it is possible to simultaneously relax the electric field concentration and reduce the increase in the electrostatic capacitance, similar to the first embodiment. In addition, because the UV light is irradiated while the mold **230** is pressed against the uncured film **201A** of the UV curable resin, the opening **210** can be easily formed with a high accuracy.

The semiconductor device **200** including the GaN-HEMT having the finger gate structure can also be manufactured by performing three exposures with respect to the photoresist film **101**, in a modification of the first embodiment. That is, the semiconductor device **200** can be manufactured by adding one exposure for forming the side surface **214** having a different inclination direction.

On the other hand, in the second embodiment, the opening **210** can be formed by a single irradiation of the UV light, using the mold **230**. Hence, according to the second embodiment, the GaN-HEMT having the finger gate structure can be easily formed by a small number of processes or steps. The method for forming the opening **210** as in the second embodiment may sometimes be referred to as a UV nano-imprint lithography method.

Next, an example of a method for forming the mold **230** will be described. FIG. **28** through FIG. **34** are cross sectional views illustrating a first example of the method for forming the mold **230**. In this example, the mold **230** is formed from quartz.

First, as illustrated in FIG. **28**, a flat quartz plate **251** is prepared, and a positive photoresist film **252** is formed on one surface **251A** of the quartz plate **251**.

Next, as illustrated in FIG. **29**, a photosensitive region **261** is formed in the photoresist film **252** by exposure. The exposure is performed by reduction exposure, for example. When performing the exposure, light **270** is irradiated from a direction inclined from a direction perpendicular to a surface **251A** of the quartz plate **251**. The photosensitive region **261** has side surfaces **271** and **272** that are parallel to each other in the cross sectional view. As described above, there are two kinds of inclination directions of the side surface **234** of the protrusion **236**. The side surface **271** is formed to correspond to the side surface **234** having one of the two kinds of inclinations. The side surface **272** is formed at a position closer to the side surface **271** than the side surface **234** having the other of the two kinds of inclinations is to the side surface **271**.

Next, as illustrated in FIG. **30**, a photosensitive region **262** partially overlapping the photosensitive region **261** is formed in the photoresist film **252** by exposure. The exposure is performed by reduction exposure, for example. When performing the exposure, light **280** is irradiated from a direction inclined from the direction perpendicular to the surface **251A** of the quartz plate **251** in a direction opposite to the direction in which the light **270** is inclined. The photosensitive region **262** is formed such that the side surface **271** remains, but the side surface **272** is incorporated into the photosensitive region **262**. The photosensitive region **262** has a side surface **281** located at a position on an outer side of the photosensitive region **261** than the position where the side surface **272** existed. The side surface **281** is formed so as to correspond to the side surface **234** having the other of the two kinds of inclinations.

A photosensitive region **260** is formed from the photosensitive regions **261** and **262** partially overlapping each

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other. The photosensitive region **260** has side surfaces **271** and **281**. The photosensitive region **260** has a cross section having an isosceles trapezoid shape.

Next, as illustrated in FIG. **31**, a photosensitive region **265** is formed in a portion between mutually adjacent photosensitive regions **260** of the photoresist film **252** by exposure. The exposure is performed by reduction exposure, for example. When performing the exposure, light **290** is irradiated from a direction perpendicular to the surface **251A** of the quartz plate **251**. The photosensitive region **265** has side surfaces **291** and **292** that are parallel to each other in the cross sectional view. The side surfaces **291** and **292** are perpendicular to surface **251A**. The side surfaces **291** and **292** are formed to correspond to the side surfaces **232** of the protrusions **236**. For example, the side surface **291** is formed so that an unexposed region of the photoresist film **252** remains between the side surface and the side surface **271**, and the side surface **292** is formed so that an unexposed region of the photoresist film **252** remains between the side surface **281** and the side surface **292**. The photosensitive region **265** has a cross section having a rectangular shape.

Next, as illustrated in FIG. **32**, the photoresist film **252** is developed, to remove the photosensitive regions **260** and **265**. As a result, an opening **243** having side surfaces **241** and **242**, and an opening **248** having side surfaces **246** and **247**, are formed in the photoresist film **252**. The side surface **241** is formed at the position of the side surface **281** of the photosensitive region **260**, and the side surface **242** is formed at the position of the side surface **271** of the photosensitive region **260**. The side surface **246** is formed at the position of the side surface **291** of the photosensitive region **265**, and the side surface **247** is formed at the position of the side surface **292** of the photosensitive region **265**. The angle formed by the surface **251A** and the side surfaces **241** and **242** may be in a range greater than or equal to 45° and less than or equal to 60° , for example, and the angle formed by the surface **251A** and the side surfaces **246** and **247** may be in a range greater than or equal to 85° and less than or equal to 90° , for example. An alkaline developer is used to develop the photoresist film **252**, for example.

Next, as illustrated in FIG. **33**, the base **235** and the plurality of protrusions **236** are formed on the quartz plate **251**, by etching the quartz plate **251** through the openings **243** and **248**. The protrusion **236** has the side surface **232** continuous with the side surface **246** or **247**, and the side surface **234** continuous with the side surface **241** or **242**. RIE is used for the etching described above, for example.

Next, as illustrated in FIG. **34**, the photoresist film **252** is removed. The photoresist film **252** can be removed by using an organic solvent, oxygen plasma, or the like, for example. Hence, the mold **230** made of quartz can be formed in the above described manner.

Next, a second example of the method for forming the mold **230** will be described. FIG. **35** through FIG. **39** are cross sectional views illustrating the second example of the method for forming the mold **230**. In this example, the mold **230** is formed of a resin.

First, as illustrated in FIG. **35**, a master mold **330** made of quartz is formed. The master mold **330** includes a flat base **335**, and a plurality of protrusions **336** protruding from the base **335**. The protrusions **336** protrude from one surface **335A** of the base **335**. The base **335** and the protrusions **336** have shapes similar to the shapes of the base **235** and the protrusions **236**, respectively. The protrusion **336** has a surface **333** parallel to the surface **335A**, a side surface **332**, and a side surface **334**. The master mold **330** has a shape

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corresponding to a repeating unit including two protrusions **236** that are adjacent to each other in the second direction of the mold **230**.

Next, as illustrated in FIG. **36**, a transfer member **340** is formed on the master mold **330**. The transfer member **340** is a nickel (Ni) film, for example. The transfer member **340** can be formed by nickel electroforming. The transfer member **340** is formed with a surface **341** having concavo-convexes conforming to the protrusions **336** of the master mold **330**.

Next, as illustrated in FIG. **37**, the transfer member **340** is released or parted from the master mold **330**.

Next, as illustrated in FIG. **38**, an uncured resin film **237** is prepared, and the transfer member **340** is pressed against one surface **237A** of the resin film **237**. As a result, concavo-convexes conforming to the surface **341** of the transfer member **340** are formed on the surface **237A**. That is, the concavo-convexes conforming to the protrusions **336** of the master mold **330** are transferred to the surface **237A**. The resin film **237** is a thermoplastic resin film or a UV curable resin film, for example. The resin film **237** may include a fluororesin, for example.

Thereafter, as illustrated in FIG. **39**, the transfer of the concavo-convexes conforming to the protrusions **336** of the master mold **330** using the transfer member **340** is repeated. Then, after all of the concavo-convexes that become the protrusions **236** are formed, the resin film **237** is cured. The resin film **237** is heated in the case where the resin film **237** is the thermoplastic resin film, and UV light irradiation is performed in the case where the resin film **237** is the UV curable resin film. Hence, the resin mold **230** can be formed in the above described manner.

According to the present disclosure, it is possible to simultaneously relax the electric field concentration and reduce an increase in the electrostatic capacitance.

While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

Although the embodiment has been described in detail above, the embodiment is not limited to a specific embodiment, and various modifications and changes can be made within the scope described in the claims.

What is claimed is:

1. A method for manufacturing a semiconductor device, comprising:

forming a semiconductor layer above a substrate;
forming a source electrode and a drain electrode on the semiconductor layer;

forming a first insulating film covering a surface of the semiconductor layer, between the source electrode and the drain electrode;

forming a second insulating film on the first insulating film;

forming a mask on the second insulating film, the mask having an opening between the source electrode and the drain electrode in a plan view viewed in a direction perpendicular to an upper surface of the substrate;

forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating

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film, by etching the first insulating film and the second insulating film through the opening; and

forming a gate electrode on the first insulating film and the second insulating film, the gate electrode making a Schottky contact with the semiconductor layer through the first gate opening and the second gate opening, wherein:

the opening includes

a first side surface, and

a second side surface closer to the drain electrode than the first side surface is to the drain electrode,

an angle formed between the first side surface and the upper surface is larger than an angle formed between the second side surface and the upper surface,

an etching rate of the second insulating film is higher than an etching rate of the first insulating film during the etching of the forming the first gate opening, and

a distance between the first gate opening and the second gate opening on a side closer to the drain electrode than the first gate opening is to the drain electrode is longer than a distance between the first gate opening and the second gate opening on a side closer to the source electrode than the first gate opening is to the source electrode.

2. The method for manufacturing the semiconductor device as claimed in claim 1, wherein

the first insulating film is a first silicon nitride film having a first refractive index, and

the second insulating film is a second silicon nitride film having a second refractive index lower than the first refractive index.

3. The method for manufacturing the semiconductor device as claimed in claim 1, wherein

the forming the mask includes

forming a positive photosensitive film on the second insulating film,

exposing the photosensitive film to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and

removing the photosensitive region by developing the photosensitive film, and

the exposing the photosensitive film includes an exposure from a direction parallel to the first side surface, and exposure from a direction parallel to the second side surface.

4. The method for manufacturing the semiconductor device as claimed in claim 1, wherein the forming the mask includes

forming a positive photosensitive film on the second insulating film,

exposing the photosensitive film by a single exposure using a shifted telecentric optical system, to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and removing the photosensitive region by developing the photosensitive film.

5. The method for manufacturing the semiconductor device as claimed in claim 1, wherein the forming the mask includes

planarizing an upper surface of the second insulating film, forming an uncured film of a resin on the planarized second insulating film,

forming a cured film by curing the uncured film while pressing a mold against the uncured film, and removing the mold from the cured film, and

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the mold includes a base, and a protrusion protruding from the base and having a shape corresponding to a shape of the opening.

6. The method for manufacturing a semiconductor device as claimed in claim 5, wherein

the resin is an ultraviolet curable resin, and the forming the cured film includes irradiating ultraviolet light on the uncured film through the mold.

7. The method for manufacturing the semiconductor device as claimed in claim 5, wherein

the source electrode and the drain electrode extend in a first direction parallel to the upper surface, and are alternately formed at multiple locations in a second direction parallel to the upper surface and perpendicular to the first direction, and

the gate electrode is formed at each location between the source electrode and the drain electrode that are adjacent to each other in the second direction.

8. A method for manufacturing a semiconductor device, comprising:

forming a semiconductor layer above a substrate;

forming a source electrode and a drain electrode on the semiconductor layer;

forming a first insulating film covering a surface of the semiconductor layer, between the source electrode and the drain electrode;

forming a second insulating film on the first insulating film;

forming a mask on the second insulating film, the mask having an opening between the source electrode and the drain electrode in a plan view viewed in a direction perpendicular to an upper surface of the substrate;

forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first insulating film and the second insulating film through the opening; and

forming a gate electrode on the first insulating film and the second insulating film, the gate electrode making a Schottky contact with the semiconductor layer through the first gate opening and the second gate opening, wherein the opening includes

a first side surface, and

a second side surface closer to the drain electrode than the first side surface is to the drain electrode,

an angle formed between the first side surface and the upper surface is larger than an angle formed between the second side surface and the upper surface,

an etching rate of the second insulating film is higher than an etching rate of the first insulating film during the etching of the forming the first gate opening,

an angle between the first side surface and the upper surface is in a range greater than or equal to 85° and less than or equal to 90°, and

an angle between the second side surface and the upper surface is in a range greater than or equal to 45° and less than or equal to 60°.

9. The method for manufacturing the semiconductor device as claimed in claim 8, wherein

the first insulating film is a first silicon nitride film having a first refractive index, and

the second insulating film is a second silicon nitride film having a second refractive index lower than the first refractive index.

10. The method for manufacturing the semiconductor device as claimed in claim 8, wherein the forming the mask includes

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forming a positive photosensitive film on the second insulating film,

exposing the photosensitive film to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and

removing the photosensitive region by developing the photosensitive film, and

the exposing the photosensitive film includes an exposure from a direction parallel to the first side surface, and exposure from a direction parallel to the second side surface.

11. The method for manufacturing the semiconductor device as claimed in claim 8, wherein the forming the mask includes

forming a positive photosensitive film on the second insulating film,

exposing the photosensitive film by a single exposure using a shifted telecentric optical system, to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and

removing the photosensitive region by developing the photosensitive film.

12. A method for manufacturing a semiconductor device, comprising:

forming a semiconductor layer above a substrate;

forming a source electrode and a drain electrode on the semiconductor layer;

forming a first insulating film covering a surface of the semiconductor layer, between the source electrode and the drain electrode;

forming a second insulating film on the first insulating film;

forming a mask on the second insulating film, the mask having an opening between the source electrode and the drain electrode in a plan view viewed in a direction perpendicular to an upper surface of the substrate;

forming a first gate opening in the first insulating film and forming a second gate opening in the second insulating film, by etching the first insulating film and the second insulating film through the opening; and

forming a gate electrode on the first insulating film and the second insulating film, the gate electrode making a Schottky contact with the semiconductor layer through the first gate opening and the second gate opening, wherein:

the opening includes

a first side surface, and

a second side surface closer to the drain electrode than the first side surface is to the drain electrode,

an angle formed between the first side surface and the upper surface is larger than an angle formed between the second side surface and the upper surface,

an etching rate of the second insulating film is higher than an etching rate of the first insulating film during the etching of the forming the first gate opening, and

an etching rate of the second insulating film is four times or more higher than an etching rate of the first insulating film during the etching of the forming the first gate opening.

13. The method for manufacturing the semiconductor device as claimed in claim 12, wherein

the first insulating film is a first silicon nitride film having a first refractive index, and

the second insulating film is a second silicon nitride film having a second refractive index lower than the first refractive index.

14. The method for manufacturing the semiconductor device as claimed in claim 12, wherein

the forming the mask includes

forming a positive photosensitive film on the second insulating film,

exposing the photosensitive film to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and

removing the photosensitive region by developing the photosensitive film, and

the exposing the photosensitive film includes an exposure from a direction parallel to the first side surface, and exposure from a direction parallel to the second side surface.

15. The method for manufacturing the semiconductor device as claimed in claim 12, wherein the forming the mask includes

forming a positive photosensitive film on the second insulating film,

exposing the photosensitive film by a single exposure using a shifted telecentric optical system, to form a photosensitive region in a portion of the photosensitive film where the opening is to be formed, and

removing the photosensitive region by developing the photosensitive film.

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